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1 Misc

1.1 Default Code [24a798]

```
1 #include <bits/stdc++.h>
2 using namespace std;
3 #define int long long
4 #define debug(a...) cerr << #a << " = ", dout(a)
5 void dout() { cerr << "\n"; }
6 template <typename A, typename... B>
7 void dout(A a, B... b) { cerr << a << ' ', dout(b...); }
8
9 void solve(){
10
11 }
12
13 signed main(){
14     ios::sync_with_stdio(0), cin.tie(0);
15
16     int t = 1;
17     while (t--){
18         solve();
19     }
20
21     return 0;
22 }
```

1.2 Run

```
1 from os import *
2 f = "pA"
3 while 1:
4     i = input("> ") or f; system("clear"); f = i
5     print(f"file = {f}")
6     if system(f"g++ {f}.cpp -std=c++17 -Wall -Wextra -Wshadow
7         -O2 -D LOCAL -g -fsanitize=undefined,address -o {f}
8         "):
9         print("CE"); continue
10    for x in sorted(listdir()):
11        if x.startswith(f) and x.endswith(".in"):
12            print(x); system(f"./{f} < {x}") and print("RE");
13            print()
```

1.3 Diff

```
1 set -e
2 g++ ac.cpp -o ac
3 g++ wa.cpp -o wa
4 for ((i=0;;i++))
5 do
6     echo "$i"
7     python3 gen.py > input
8     ./ac < input > ac.out
9     ./wa < input > wa.out
10    diff ac.out wa.out || break
11 done
```

1.4 Hash Command

```
1 cat file.cpp | cpp -dD -P -fpreprocessed | tr -d "[:space:]"
2 | md5sum | cut -c-6
```

1.5 Vimrc

```
1 se nu rnu bs=2 sw=4 ts=4 hls ls=2 si acd bo=all mouse=a
2
3 :inoremap " ""<Esc>i
4 :inoremap {<CR> {<CR><Esc>ko
5 :inoremap [{ {<Esc>i
6
7 ca hash w !cpp -dD -P -fpreprocessed \\\ tr -d "[:space:]" \\\
8 md5sum \\\ cut -c-6
9
10 " i+<esc>25A---+<esc>
11 " o/<esc>25A |<esc>
12 " ggVGyG35pGdd
```

1.6 Custom Set PQ Sort [d4df55]

```
1 // 所有自訂的結構體，務必檢查相等的 case，給所有元素一個排序
2 // 的依據
3 struct my_struct{
4     int val;
5     my_struct(int _val) : val(_val) {}
6 };
7
8 auto cmp = [](my_struct a, my_struct b) {
9     return a.val > b.val;
10 };
11
12 set<my_struct, decltype(cmp)> ss({1, 2, 3}, cmp);
13 priority_queue<my_struct, vector<my_struct>, decltype(cmp)>
14     pq(cmp, {1, 2, 3});
15 map<my_struct, my_struct, decltype(cmp)> mp({{1, 4}, {2, 5},
16     {3, 6}}, cmp);
```

1.7 Dynamic Bitset [c78aa8]

```
1 const int MAXN = 2e5 + 5;
2 template <int len = 1>
3 void solve(int n) {
4     if (n > len) {
5         solve<min(len*2, MAXN)>(n);
6         return;
7     }
8     bitset<len> a;
9 }
```

1.8 Enumerate Subset [a13e46]

```
1 // 時間複雜度 O(3^n)
2 // 枚舉每個 mask 的子集
3 for (int mask=0; mask<(1<n); mask++){
4     for (int s=mask; s>=0; s=(s-1)&m){
5         // s 是 mask 的子集
6         if (s==0) break;
7     }
8 }
```

1.9 Fast Input [6f8879]

```
1 // fast IO
2 // 6f8879
3 inline char readchar(){
```

```
4     static char buffer[BUFSIZ], *now = buffer + BUFSIZ, *
5     end = buffer + BUFSIZ;
6     if (now == end)
7     {
8         if (end < buffer + BUFSIZ)
9             return EOF;
10        end = (buffer + fread(buffer, 1, BUFSIZ, stdin));
11        now = buffer;
12    }
13    return *now++;
14 }
15 inline int nextint(){
16     int x = 0, c = readchar(), neg = false;
17     while(('0' > c || c > '9') && c!='-' && c!=EOF) c =
18         readchar();
19     if(c == '-') neg = true, c = readchar();
20     while('0' <= c && c <= '9') x = (x<<3) + (x<<1) + (c^'0')
21         , c = readchar();
22     if(neg) x = -x;
23     return x; // returns 0 if EOF
24 }
```

1.10 OEIS [25cf74]

```
1
2 // 若一個線性遞迴有 k 項，給他恰好 2*k 個項可以求出線性遞迴
3 // s = 1 1 3 7 17，則它會回傳 1 2
4 vector<int> BerlekampMassey(vector<int> s) {
5     if (s.empty()) return vector<int>();
6     int n = s.size(), L = 0, m = 0;
7     vector<int> C(n), B(n), T;
8     C[0] = B[0] = 1;
9
10    int b = 1;
11    for (int i=0; i<n; i++) {
12        ++m; int d = s[i]%MOD;
13        for (int j=1; j<L+1; j++) d = (d+C[j]*s[i-j])%MOD;
14        if (!d) continue;
15        T = C; int coef = d*qp(b, MOD-2, MOD)%MOD;
16        for (int j=m; j<n; j++) C[j] = (C[j]-coef*B[j-m])%
17            MOD;
18        if (2*L>i) continue;
19        L = i+1-L; B=T; b=d; m=0;
20    }
21    C.resize(L+1); C.erase(C.begin());
22    for (int &x : C) x = (MOD-x)%MOD;
23    reverse(C.begin(), C.end());
24    return C;
25 }
```

1.11 OEIS

```
1 from fractions import Fraction as F
2 def BM(s):
3     n = len(s); L=m=0; C=[F(0)]*n; B=[F(0)]*n; C[0]=B[0]=F(1)
4     ; b=F(1)
5     for i in range(n):
6         m+=1; d=sum(C[j]*s[i-j] for j in range(L+1))
7         if d==0: continue
8         T = C[:]; coef = F(d/b)
9         for j in range(m, n): C[j] -= coef*B[j-m]
10        if 2*L<=i: L, B, b, m = i+1-L, T, d, 0
11    if len(C)<L+1: C += [F(0)] * (L+1-len(C))
12    return [-x for x in C[1:L+1]]
```

1.12 Pragma [09d13e]

```
1 #pragma GCC optimize("O3,unroll-loops")
2 #pragma GCC target("avx,avx2,sse,sse2,sse3,sse4,popcnt")
```

1.13 Python

```
1 # Decimal
2 from decimal import *
3 getcontext().prec = 6
4
5 # system setting
6 sys.setrecursionlimit(100000)
7 sys.set_int_max_str_digits(10000)
8
9 # turtle
10 from turtle import *
11
12 N = 3000000010
13 setworldcoordinates(-N, -N, N, N)
14 hideturtle()
15 speed(100)
16
17 def draw_line(a, b, c, d):
18     teleport(a, b)
19     goto(c, d)
20
21 def write_dot(x, y, text, diff=1): # diff = 文字的偏移
22     teleport(x, y)
23     dot(5, "red")
24
25     teleport(x+N/100*diff, y+N/100*diff)
26     write(text, font=("Arial", 5, "bold"))
27
28 # usage
29 draw_line(*a[i], *(a[i-1]))
30 write_dot(*a[i], str(a[i]))
31 # 以自己左方 100 單位為圓心向前畫一個 90 度的弧 / 正五邊形
   ( 參數可以是負的 )
32 circle(100, extent = 90)
33 circle(100, steps = 5)
34 left(90) # 左轉 90 度
35 fd(-100) # 倒退 100 單位
36 penup()
37 pendown()
38
39 tracer(0, 0) # IO 優化 · 必須在程式碼最後手動 update()
40 pos() # 回傳目前座標
41 heading() # 回傳目前方向：0 是右方、90 是上方
42 color("red") # 改變顏色 · 可以用文字或是 #000000 的格式
43
44 # OOP
45 class Point:
46     def __init__(self, x, y):
47         self.x = x
48         self.y = y
49
50     def __add__(self, o): # use dir(int) to know operator
       name
51         return Point(self.x+o.x, self.y+o.y)
52
53 @property
54 def distance(self):
55     return (self.x**2 + self.y**2)**(0.5)
```

```
56
57 a = Point(3, 4)
58 print(a.distance)
59
60 # Fraction
61 from fractions import Fraction
62 a = Fraction(Decimal(1.1))
63 a.numerator # 分子
64 a.denominator # 分母
```

1.14 Xor Basis [840136]

```
1 vector<int> basis;
2 void add_vector(int x){
3     for (auto v : basis){
4         x=min(x, x^v);
5     }
6     if (x) basis.push_back(x);
7 }
8
9 // 給一數字集合 S · 求能不能 XOR 出 x
10 bool check(int x){
11     for (auto v : basis){
12         x=min(x, x^v);
13     }
14     return 0;
15 }
16
17 // 給一數字集合 S · 求能 XOR 出多少數字
18 // 答案等於 2^{basis 的大小}
19
20 // 給一數字集合 S · 求 XOR 出最大的數字
21 int get_max(){
22     int ans=0;
23     for (auto v : basis){
24         ans=max(ans, ans^v);
25     }
26     return ans;
27 }
```

1.15 disable ASLR [29a49f]

```
1 // Disable randomization of memory addresses
2 // setarch $(uname -m) -R /bin/bash
3
4 #include <sys/personality.h>
5 #include <bits/stdc++.h>
6 int main(int argc, char* argv[]){
7     personality(ADDR_NO_RANDOMIZE);
8     execv(argv[1], argv+1);
9 }
```

1.16 hash windows

```
1 def get_hash(path):
2     from subprocess import run, PIPE
3     from hashlib import md5
4
5     p = run(
6         ["cpp", "-dD", "-P", "-fpreprocessed", path],
7         stdout = PIPE,
8         stderr = PIPE,
9         text = True
```

```
10     )
11
12     if p.returncode != 0:
13         raise RuntimeError(p.stderr)
14
15     s = ''.join(p.stdout.split())
16     ret = md5(s.encode()).hexdigest()
17     return ret[:6]
18
19 print(get_hash("Suffix_Array.cpp"))
```

1.17 random int [9cc603]

```
1 mt19937 seed(chrono::steady_clock::now().time_since_epoch().
   count());
2 int rng(int l, int r){
3     return uniform_int_distribution<int>(l, r)(seed);
4 }
```

2 Convolution

2.1 FFT [16591a]

```
1 typedef complex<double> cd;
2
3 // 778272
4 void FFT(vector<cd> &a) {
5     int n = a.size(), L = 31-__builtin_clz(n);
6     vector<complex<long double>> R(2, 1);
7     vector<cd> rt(2, 1);
8     for (int k=2 ; k<n ; k*=2){
9         R.resize(n), rt.resize(n);
10        auto x = polar(1.0L, acos(-1.0L) / k);
11        for (int i=k ; i<2*k ; i++) rt[i] = R[i] = (i&1 ? R[i
12            /2]*x : R[i/2]);
13    }
14
15    vector<int> rev(n);
16    for (int i=0 ; i<n ; i++) rev[i] = (rev[i/2] | (i&1)<<L)
17        /2;
18    for (int i=0 ; i<n ; i++) if (i<rev[i]) swap(a[i], a[rev[
19        i]]);
20    for (int k=1 ; k<n ; k*=2){
21        for (int i=0 ; i<n ; i+=2*k){
22            for (int j=0 ; j<k ; j++){
23                // cd z = rt[j+k] * a[i+j+k];
24                auto x = (double *)&rt[j+k], y = (double *)&a
25                    [i+j+k];
26                cd z(x[0]*y[0] - x[1]*y[1], x[0]*y[1] + x[1]*
27                    y[0]);
28                a[i+j+k] = a[i+j]-z;
29                a[i+j] += z;
30            }
31        }
32    }
33 }
34
35 // 192528
36 vector<double> PolyMul(const vector<double> a, const vector<
   double> b){
37     if (a.empty() || b.empty()) return {};
38     vector<double> res(a.size()+b.size()-1);
39     int L = 32 - __builtin_clz(res.size()), n = 1<<L;
40     vector<cd> in(n), out(n);
41     copy(a.begin(), a.end(), in.begin());
```

```

37 for (int i=0 ; i<b.size() ; i++) in[i].imag(b[i]);
38 FFT(in);
39 for (cd& x : in) x *= x;
40 for (int i=0 ; i<n ; i++) out[i] = in[-i & (n - 1)] -
    conj(in[i]);
41 FFT(out);
42 for (int i=0 ; i<res.size() ; i++) res[i] = imag(out[i])
    / (4 * n);
43 return res;
44 }

```

2.2 FFT any mod [412000]

```

1 // n=524288, ~400ms
2 const int MOD = 998244353;
3 typedef complex<double> cd;
4
5 // bb6d94
6 void FFT(vector<cd> &a) {
7     int n = a.size(), L = 31-__builtin_clz(n);
8     vector<complex<long double>> R(2, 1);
9     vector<cd> rt(2, 1);
10    for (int k=2 ; k<n ; k*=2){
11        R.resize(n);
12        rt.resize(n);
13        auto x = polar(1.0L, acos(-1.0L) / k);
14        for (int i=k ; i<2*k ; i++){
15            rt[i] = R[i] = (i&1 ? R[i/2]*x : R[i/2]);
16        }
17    }
18
19    vector<int> rev(n);
20    for (int i=0 ; i<n ; i++) rev[i] = (rev[i/2] | (i&1)<<L)
        /2;
21    for (int i=0 ; i<n ; i++) if (i<rev[i]) swap(a[i], a[rev[
        i]]);
22    for (int k=1 ; k<n ; k*=2){
23        for (int i=0 ; i<n ; i+=2*k){
24            for (int j=0 ; j<k ; j++){
25                auto x = (double *)&rt[j+k];
26                auto y = (double *)&a[i+j+k];
27                cd z(x[0]*y[0] - x[1]*y[1], x[0]*y[1] + x[1]*
                    y[0]);
28                a[i+j+k] = a[i+j]-z;
29                a[i+j] += z;
30            }
31        }
32    }
33 }
34 }
35
36 // c3dcf6
37 vector<int> PolyMul(vector<int> a, vector<int> b){
38     if (a.empty() || b.empty()) return {};
39     vector<int> res(a.size()+b.size()-1);
40     int B = 32-__builtin_clz(res.size()), n = (1<<B), cut = (
        int)(sqrt(MOD));
41     vector<cd> L(n), R(n), outs(n), outl(n);
42
43     for (int i=0 ; i<a.size() ; i++) L[i] = cd((int) a[i]/cut
        , (int)a[i]%cut);
44     for (int i=0 ; i<b.size() ; i++) R[i] = cd((int) b[i]/cut
        , (int)b[i]%cut);
45     FFT(L), FFT(R);
46     for (int i=0 ; i<n ; i++){

```

```

47         int j = -i&(n-1);
48         outl[j] = (L[i]+conj(L[j])) * R[i]/(2.0*n);
49         outs[j] = (L[i]-conj(L[j])) * R[i]/(2.0*n)/1i;
50     }
51     FFT(outl), FFT(outs);
52     for (int i=0 ; i<res.size() ; i++){
53         int av = (int)(real(outl[i])+0.5), cv = (int)(imag(
            outs[i])+0.5);
54         int bv = (int)(imag(outl[i])+0.5) + (int)(real(outs[i
            ])+0.5);
55         res[i] = ((av%MOD*cut+bv) % MOD*cut+cv) % MOD;
56     }
57     return res;
58 }

```

2.3 FWT [e50788]

```

1 // 已經把 mint 刪掉 · 需要增加註解
2 vector<int> xor_convolution(vector<int> a, vector<int> b, int
    k) {
3     if (k==0) return vector<int>{a[0]*b[0]};
4     vector<int> aa(1 << (k - 1)), bb(1 << (k - 1));
5     for (int i=0 ; i<(1<<(k-1)) ; i++){
6         aa[i] = a[i] + a[i + (1 << (k - 1))];
7         bb[i] = b[i] + b[i + (1 << (k - 1))];
8     }
9     vector<int> X = xor_convolution(aa, bb, k - 1);
10    for (int i=0 ; i<(1<<(k-1)) ; i++){
11        aa[i] = a[i] - a[i + (1 << (k - 1))];
12        bb[i] = b[i] - b[i + (1 << (k - 1))];
13    }
14    vector<int> Y = xor_convolution(aa, bb, k - 1);
15    vector<int> c(1 << k);
16    for (int i=0 ; i<(1<<(k-1)) ; i++){
17        c[i] = (X[i] + Y[i]) / 2;
18        c[i + (1 << (k - 1))] = (X[i] - Y[i]) / 2;
19    }
20    return c;
21 };

```

2.4 Min Convolution Concave Concave [ffb28d]

```

1 // 需要增加註解
2 // min convolution
3 vector<int> mkk(vector<int> a, vector<int> b) {
4     vector<int> slope;
5     FOR (i, 1, ssize(a) - 1) slope.pb(a[i] - a[i - 1]);
6     FOR (i, 1, ssize(b) - 1) slope.pb(b[i] - b[i - 1]);
7     sort(all(slope));
8     slope.insert(begin(slope), a[0] + b[0]);
9     partial_sum(all(slope), begin(slope));
10    return slope;
11 }

```

2.5 NTT mod 998244353 [976c8d]

```

1
2 const int MOD = (119 << 23) + 1, ROOT = 62; // = 998244353
3 // For p < 2^30 there is also e.g. 5 << 25, 7 << 26, 479 <<
    21
4 // and 483 << 21 (same root). The last two are > 10^9.
5
6 // 7e041b
7 vector<int> rt(2, 1);

```

```

8 void NTT_init(int n){
9     static bool lastInit = 0;
10    if (lastInit<n){
11        lastInit = n;
12        for (int k=2, s=2 ; k<n ; k*=2, s++){
13            rt.resize(n);
14            int z[] = {1, qp(ROOT, MOD>>s, MOD)};
15            for (int i=k ; i<2*k ; i++) rt[i] = rt[i/2]*z[i
                &1]%MOD;
16        }
17    }
18 }
19
20 // fec4b8
21 void NTT(vector<int> &a, int isInverse = false) {
22     int n = a.size();
23     int L = 31-__builtin_clz(n);
24     NTT_init(n);
25
26     vector<int> rev(n);
27     for (int i=0 ; i<n ; i++) rev[i] = (rev[i/2] | (i&1)<<L)/2;
28     for (int i=0 ; i<n ; i++){
29         if (i<rev[i]) swap(a[i], a[rev[i]]);
30     }
31
32     for (int k=1 ; k<n ; k*=2){
33         for (int i=0 ; i<n ; i+=2*k){
34             for (int j=0 ; j<k ; j++){
35                 int z = rt[j+k]*a[i+j+k]%MOD, &ai = a[i+j];
36                 a[i+j+k] = ai-z+(z>ai ? MOD : 0);
37                 ai += (ai+z)>MOD ? z-MOD : z;
38             }
39         }
40     }
41
42     if (isInverse) {
43         reverse(a.begin()+1, a.end());
44         int n_inv = qp(n, MOD-2, MOD);
45         for (int i=0 ; i<n ; i++) a[i] = a[i]*n_inv%MOD;
46     }
47 }
48
49 // 771576
50 vector<int> polyMul(vector<int> a, vector<int> b){
51     if (a.empty() || b.empty()) return {};
52     int s = a.size()+b.size()-1, B = 32-__builtin_clz(s), n =
        1<<B;
53     vector<int> L(a), R(b), out(n);
54     L.resize(n), R.resize(n);
55     NTT(L), NTT(R);
56     for (int i=0 ; i<n ; i++) out[i] = L[i]*R[i]%MOD;
57     NTT(out, true);
58     out.resize(s);
59     return out;
60 }

```

2.6 PolyInv [b388fb]

```

1 vector<int> polyInv(vector<int> a){
2     int n = a.size();
3     vector<int> ret(1, qp(a[0], MOD-2, MOD));
4
5     for (int m=1 ; m<n ; m<=1){
6         if (n<2*m) a.resize(2*m);
7

```

```

8   vector<int> v1(a.begin(), a.begin()+2*m), v2 = ret;
9   v1.resize(4*m), v2.resize(4*m);
10  NTT(v1), NTT(v2);
11  for (int i=0; i<4*m; i++) v1[i] = v1[i]*v2[i]%MOD*
    v2[i]%MOD;
12  NTT(v1, true);
13  ret.resize(2*m);
14  for (int i=0; i<m; i++) ret[i] = (ret[i]+ret[i])%
    MOD;
15  for (int i=0; i<2*m; i++) ret[i] = ((ret[i]-v1[i])%
    MOD+MOD)%MOD;
16  }
17  ret.resize(n);
18  return ret;
19 }

```

2.7 PolySqrt [1ea321]

```

1  // ---- sqrt for formal power series over MOD=998244353 ----
2  int mod_sqrt(int a) { // Tonelli-Shanks; return -1 if no
    solution
3  a %= MOD; if (a < 0) a += MOD;
4  if (a == 0) return 0;
5  if (qp(a, (MOD - 1) / 2, MOD) == MOD - 1) return -1; //
    non-residue
6  if (MOD % 4 == 3) return qp(a, (MOD + 1) / 4, MOD);
7  int q = MOD - 1, s = 0;
8  while ((q & 1) == 0) q >>= 1, ++s;
9  int z = 2;
10 while (qp(z, (MOD - 1) / 2, MOD) != MOD - 1) ++z; // find
    non-residue
11 long long c = qp(z, q, MOD);
12 long long x = qp(a, (q + 1) / 2, MOD);
13 long long t = qp(a, q, MOD);
14 int m = s;
15 while (t != 1) {
16     int i = 1;
17     long long tt = t * t % MOD;
18     while (tt != 1) {
19         tt = tt * tt % MOD;
20         ++i;
21         if (i == m) return -1; // shouldn't happen
22     }
23     long long b = qp(c, 1LL << (m - i - 1), MOD);
24     x = x * b % MOD;
25     long long b2 = b * b % MOD;
26     t = t * b2 % MOD;
27     c = b2;
28     m = i;
29 }
30 return (int)x;
31 }
32 }

```

```

33 vector<int> polySqrt(vector<int> a) {
34     int n = (int)a.size();
35
36     // all zero -> zero
37     bool all0 = true;
38     for (int x : a) if (x % MOD != 0) { all0 = false; break;
39     }
40     if (all0) return vector<int>(n, 0);
41
42     // first nonzero position
43     int k = 0;

```

```

44 while (k < n && a[k] == 0) ++k;
45 if (k & 1) return {}; // odd -> impossible
46 int shift = k / 2;
47
48 // strip x^k
49 vector<int> f;
50 f.reserve(n - k);
51 for (int i = k; i < n; ++i) f.push_back(a[i]);
52
53 // constant term sqrt, choose canonical sign
54 int c0 = (f.empty() ? 0 : f[0] % MOD);
55 int s0 = mod_sqrt(c0);
56 if (s0 == -1) return {};
57 if (s0 > MOD - s0) s0 = MOD - s0; // choose the smaller
    nonnegative root
58
59 // We must produce N terms after placing the shift =>
    need = N - shift
60 int need = n - shift;
61 if ((int)f.size() < need) f.resize(need, 0);
62
63 // Newton: g_{new} = (g + f/g) / 2 (mod x^m)
64 const int INV2 = (MOD + 1) / 2;
65 vector<int> g(1, s0);
66
67 for (int m = 1; m < need; m <= 1) {
68     int lim = min(need, m << 1);
69
70     vector<int> g_cut = g; g_cut.resize(lim);
71     vector<int> inv_g = polyInv(g_cut); // inv up
    to lim
72     vector<int> f_cut(f.begin(), f.begin() + lim);
73
74     vector<int> q = polyMul(inv_g, f_cut);
75     if ((int)q.size() > lim) q.resize(lim);
76
77     g.resize(lim);
78     for (int i = 0; i < lim; ++i) {
79         long long gi = (i < (int)g_cut.size() ? g_cut[i]
80             : 0);
81         long long qi = (i < (int)q.size() ? q[i] : 0);
82         g[i] = (gi + qi) % MOD * 1LL * INV2 % MOD;
83     }
84     g.resize(need);
85
86     // place back the x^{k/2} shift
87     vector<int> res(n, 0);
88     for (int i = 0; i < need && i + shift < n; ++i) res[i +
    shift] = g[i];
89     return res;
90 }

```

3 Data-Structure

3.1 BIT [7ef3a9]

```

1 vector<int> BIT(MAX_SIZE);
2
3 // const int MAX_N = (1<<20)
4 int k_th(int k) { // 回傳 BIT 中第 k 小的元素 (based-1)
5     int res = 0;
6     for (int i=MAX_N>>1; i>=1; i>>=1)
7         if (BIT[res+i]<k)
8             k -= BIT[res+i];

```

```

9     return res+1;
10 }

```

3.2 Disjoint Set Persistent [447002]

```

1 struct Persistent_Disjoint_Set{
2     Persistent_Segment_Tree arr, sz;
3
4     void init(int n){
5         arr.init(n);
6         vector<int> v1;
7         for (int i=0; i<n; i++){
8             v1.push_back(i);
9         }
10        arr.build(v1, 0);
11
12        sz.init(n);
13        vector<int> v2;
14        for (int i=0; i<n; i++){
15            v2.push_back(1);
16        }
17        sz.build(v2, 0);
18    }
19
20    int find(int a){
21        int res = arr.query_version(a, a+1, arr.version.size()
22            -1).val;
23        if (res==a) return a;
24        return find(res);
25    }
26
27    bool unite(int a, int b){
28        a = find(a);
29        b = find(b);
30
31        if (a!=b){
32            int sz1 = sz.query_version(a, a+1, arr.version.
33                size()-1).val;
34            int sz2 = sz.query_version(b, b+1, arr.version.
35                size()-1).val;
36
37            if (sz1<sz2){
38                arr.update_version(a, b, arr.version.size()
39                    -1);
40                sz.update_version(b, sz1+sz2, arr.version.
41                    size()-1);
42            }else{
43                arr.update_version(b, a, arr.version.size()
44                    -1);
45                sz.update_version(a, sz1+sz2, arr.version.
46                    size()-1);
47            }
48            return true;
49        }
50        return false;
51    }
52 }

```

3.3 PBDS GP Hash Table [866cf6]

```

1 #include <ext/pb_ds/assoc_container.hpp>
2 using namespace __gnu_pbds;
3 typedef tree<int, null_type, less<int>, rb_tree_tag,
4     tree_order_statistics_node_update> order_set;

```

```

4 struct custom_hash {
5     static uint64_t splitmix64(uint64_t x) {
6         // http://xorshift.di.unimi.it/splitmix64.c
7         x += 0x9e3779b97f4a7c15;
8         x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
9         x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
10        return x ^ (x >> 31);
11    }
12
13    size_t operator()(uint64_t x) const {
14        static const uint64_t FIXED_RANDOM = chrono::
15            steady_clock::now().time_since_epoch().count();
16        return splitmix64(x + FIXED_RANDOM);
17    };
18
19 gp_hash_table<int, int, custom_hash> ss;

```

3.4 PBDS Order Set [231774]

```

1 // .find_by_order(k) 回傳第 k 小的值 (based-0)
2 // .order_of_key(k) 回傳有多少元素比 k 小
3 // 不能在 #define int long long 後 #include 檔案
4 #include <ext/pb_ds/assoc_container.hpp>
5 #include <ext/pb_ds/tree_policy.hpp>
6 using namespace __gnu_pbds;
7 typedef tree<int, null_type, less<int>, rb_tree_tag,
8     tree_order_statistics_node_update> order_set;

```

3.5 Segment Tree Add Set [bb1898]

```

1 // [ll, rr), based-0
2 // 使用前記得 init(陣列大小), build(陣列名稱)
3 // add(ll, rr): 區間修改
4 // set(ll, rr): 區間賦值
5 // query(ll, rr): 區間求和 / 求最大值
6 struct SegmentTree{
7     struct node{
8         int add_tag = 0;
9         int set_tag = 0;
10        int sum = 0;
11        int ma = 0;
12    };
13
14    vector<node> arr;
15
16    SegmentTree(int n){
17        arr.resize(n<<2);
18    }
19
20    node pull(node A, node B){
21        node C;
22        C.sum = A.sum+B.sum;
23        C.ma = max(A.ma, B.ma);
24        return C;
25    }
26
27    // cce0c8
28    void push(int idx, int ll, int rr){
29        if (arr[idx].set_tag!=0){
30            arr[idx].sum = (rr-ll)*arr[idx].set_tag;
31            arr[idx].ma = arr[idx].set_tag;
32            if (rr-ll>1){
33                arr[idx*2+1].add_tag = 0;

```

```

34            arr[idx*2+1].set_tag = arr[idx].set_tag;
35            arr[idx*2+2].add_tag = 0;
36            arr[idx*2+2].set_tag = arr[idx].set_tag;
37        }
38        arr[idx].set_tag = 0;
39    }
40
41    if (arr[idx].add_tag!=0){
42        arr[idx].sum += (rr-ll)*arr[idx].add_tag;
43        arr[idx].ma += arr[idx].add_tag;
44        if (rr-ll>1){
45            arr[idx*2+1].add_tag += arr[idx].add_tag;
46            arr[idx*2+2].add_tag += arr[idx].add_tag;
47        }
48        arr[idx].add_tag = 0;
49    }
50
51    void build(vector<int> &v, int idx = 0, int ll = 0, int
52        rr = n){
53        if (rr-ll==1){
54            arr[idx].sum = v[ll];
55            arr[idx].ma = v[ll];
56        }else{
57            int mid = (ll+rr)/2;
58            build(v, idx*2+1, ll, mid);
59            build(v, idx*2+2, mid, rr);
60            arr[idx] = pull(arr[idx*2+1], arr[idx*2+2]);
61        }
62    }
63
64    void add(int ql, int qr, int val, int idx = 0, int ll =
65        0, int rr = n){
66        push(idx, ll, rr);
67        if (rr<=ql || qr<=ll) return;
68        if (ql<=ll && rr<=qr){
69            arr[idx].add_tag += val;
70            push(idx, ll, rr);
71            return;
72        }
73        int mid = (ll+rr)/2;
74        add(ql, qr, val, idx*2+1, ll, mid);
75        add(ql, qr, val, idx*2+2, mid, rr);
76        arr[idx]=pull(arr[idx*2+1], arr[idx*2+2]);
77    }
78
79    void set(int ql, int qr, int val, int idx=0, int ll=0,
80        int rr=n){
81        push(idx, ll, rr);
82        if (rr<=ql || qr<=ll) return;
83        if (ql<=ll && rr<=qr){
84            arr[idx].add_tag = 0;
85            arr[idx].set_tag = val;
86            push(idx, ll, rr);
87            return;
88        }
89        int mid = (ll+rr)/2;
90        set(ql, qr, val, idx*2+1, ll, mid);
91        set(ql, qr, val, idx*2+2, mid, rr);
92        arr[idx] = pull(arr[idx*2+1], arr[idx*2+2]);
93    }
94
95    node query(int ql, int qr, int idx = 0, int ll = 0, int
96        rr = n){
97        push(idx, ll, rr);
98        if (rr<=ql || qr<=ll) return node();
99        if (ql<=ll && rr<=qr) return arr[idx];

```

```

96
97        int mid = (ll+rr)/2;
98        return pull(query(ql, qr, idx*2+1, ll, mid), query(ql
99            , qr, idx*2+2, mid, rr));
100    } ST;

```

3.6 Segment Tree Li Chao Line [45b8ba]

```

1 // 全部都是 0-based
2 // 宣告: LC_Segment_Tree st(n);
3 // 函式:
4 // update({a, b}): 插入一條 y=ax+b 的全域直線
5 // query(x): 查詢所有直線在位置 x 的最小值
6 const int MAX_V = 1e6+10; // 值域最大值
7
8 struct LC_Segment_Tree{
9     struct Node{ // y = ax+b
10        int a = 0;
11        int b = INF;
12
13        int y(int x){
14            return a*x+b;
15        }
16    };
17    vector<Node> arr;
18
19    LC_Segment_Tree(int n = 0){
20        arr.resize(4*n);
21    }
22
23    void update(Node val, int idx = 0, int ll = 0, int rr =
24        MAX_V){
25        if (rr-ll==0) return;
26        if (rr-ll==1){
27            if (val.y(ll)<arr[idx].y(ll)){
28                arr[idx] = val;
29            }
30            return;
31        }
32
33        int mid = (ll+rr)/2;
34        if (arr[idx].a > val.a) swap(arr[idx], val); // 原本
35        的線斜率要比較小
36        if (arr[idx].y(mid) < val.y(mid)){ // 交點在左邊
37            update(val, idx*2+1, ll, mid);
38        }else{ // 交點在右邊
39            swap(arr[idx], val); // 在左子樹中，新線比舊線還
40            要好
41            update(val, idx*2+2, mid, rr);
42        }
43        return;
44    }
45
46    int query(int x, int idx = 0, int ll = 0, int rr = MAX_V)
47    {
48        if (rr-ll==0) return INF;
49        if (rr-ll==1){
50            return arr[idx].y(ll);
51        }
52
53        int mid = (ll+rr)/2;
54        if (x<mid){

```



```

51         return min(arr[idx].y(x), query(x, idx*2+1, ll,
52             mid));
53     }else{
54         return min(arr[idx].y(x), query(x, idx*2+2, mid,
55             rr));
56     }
57 }
58 };

```

3.7 Segment Tree Li Chao Segment [2cb0a4]

```

1 // 全部都是 0-based
2 // 宣告：LC_Segment_Tree st(n);
3 // 函式：
4 // update_segment({a, b}, ql, qr)：在 [ql, qr) 插入一條 y=ax+
5 //   b 的線段
6 // query(x)：查詢所有直線在位置 x 的最小值
7 const int MAX_V = 1e6+10; // 值域最大值
8 struct LC_Segment_Tree{
9     struct Node{ // y = ax+b
10         int a = 0;
11         int b = INF;
12
13         int y(int x){
14             return a*x+b;
15         }
16     };
17     vector<Node> arr;
18
19     LC_Segment_Tree(int n = 0){
20         arr.resize(4*n);
21     }
22
23     void update(Node val, int idx = 0, int ll = 0, int rr =
24         MAX_V){
25         if (rr-ll==0) return;
26         if (rr-ll<=1){
27             if (val.y(ll)<arr[idx].y(ll)){
28                 arr[idx] = val;
29             }
30             return;
31         }
32         int mid = (ll+rr)/2;
33         if (arr[idx].a > val.a) swap(arr[idx], val); // 原本
34         // 的線斜率要比較小
35         if (arr[idx].y(mid) < val.y(mid)){ // 交點在左邊
36             update(val, idx*2+1, ll, mid);
37         }else{ // 交點在右邊
38             swap(arr[idx], val); // 在左子樹中，新線比舊線還
39             // 要好
40             update(val, idx*2+2, mid, rr);
41         }
42         return;
43     }
44
45     // 在 [ql, qr) 加上一條 val 的線段
46     void update_segment(Node val, int ql, int qr, int idx =
47         0, int ll = 0, int rr = MAX_V){
48         if (rr-ll==0) return;
49         if (rr<=ql || qr<=ll) return;
50         if (ql<=ll && rr<=qr){

```

```

48         update(val, idx, ll, rr);
49         return;
50     }
51
52     int mid = (ll+rr)/2;
53     update_segment(val, ql, qr, idx*2+1, ll, mid);
54     update_segment(val, ql, qr, idx*2+2, mid, rr);
55     return;
56 }
57
58 int query(int x, int idx = 0, int ll = 0, int rr = MAX_V)
59 {
60     if (rr-ll==0) return INF;
61     if (rr-ll==1){
62         return arr[idx].y(ll);
63     }
64
65     int mid = (ll+rr)/2;
66     if (x<mid){
67         return min(arr[idx].y(x), query(x, idx*2+1, ll,
68             mid));
69     }else{
70         return min(arr[idx].y(x), query(x, idx*2+2, mid,
71             rr));
72     }
73 }

```

3.8 Segment Tree Persistent [3b5aa9]

```

1 // 全部都是 0-based
2 // Persistent_Segment_Tree st(n+q);
3 // st.build(v, 0);
4 // 函式：
5 // update_version(pos, val, ver)：對版本 ver 的 pos 位置改成
6 //   val
7 // query_version(ql, qr, ver)：對版本 ver 查詢 [ql, qr) 的區
8 //   間和
9 // clone_version(ver)：複製版本 ver 到最新的版本
10 struct Persistent_Segment_Tree{
11     int node_cnt = 0;
12     struct Node{
13         int lc = -1;
14         int rc = -1;
15         int val = 0;
16     };
17     vector<Node> arr;
18     vector<int> version;
19
20     Persistent_Segment_Tree(int sz){
21         arr.resize(32*sz);
22         version.push_back(node_cnt++);
23         return;
24     }
25
26     void pull(Node &c, Node a, Node b){
27         c.val = a.val+b.val;
28         return;
29     }
30
31     void build(vector<int> &v, int idx, int ll = 0, int rr =
32         n){
33         auto &now = arr[idx];
34         if (rr-ll==1){

```

```

33         now.val = v[ll];
34         return;
35     }
36
37     int mid = (ll+rr)/2;
38     now.lc = node_cnt++;
39     now.rc = node_cnt++;
40     build(v, now.lc, ll, mid);
41     build(v, now.rc, mid, rr);
42     pull(now, arr[now.lc], arr[now.rc]);
43     return;
44 }
45
46 void update(int pos, int val, int idx, int ll = 0, int rr
47     = n){
48     auto &now = arr[idx];
49
50     if (rr-ll==1){
51         now.val = val;
52         return;
53     }
54
55     int mid = (ll+rr)/2;
56     if (pos<mid){
57         arr[node_cnt] = arr[now.lc];
58         now.lc = node_cnt;
59         node_cnt++;
60         update(pos, val, now.lc, ll, mid);
61     }else{
62         arr[node_cnt] = arr[now.rc];
63         now.rc = node_cnt;
64         node_cnt++;
65         update(pos, val, now.rc, mid, rr);
66     }
67     pull(now, arr[now.lc], arr[now.rc]);
68     return;
69 }
70
71 void update_version(int pos, int val, int ver){
72     update(pos, val, version[ver]);
73 }
74
75 Node query(int ql, int qr, int idx, int ll = 0, int rr =
76     n){
77     auto &now = arr[idx];
78
79     if (ql<=ll && rr<=qr) return now;
80     if (rr<=ql || qr<=ll) return Node();
81
82     int mid = (ll+rr)/2;
83
84     Node ret;
85     pull(ret, query(ql, qr, now.lc, ll, mid), query(ql,
86         qr, now.rc, mid, rr));
87     return ret;
88 }
89
90 Node query_version(int ql, int qr, int ver){
91     return query(ql, qr, version[ver]);
92 }
93
94 void clone_version(int ver){
95     version.push_back(node_cnt);
96     arr[node_cnt] = arr[version[ver]];
97     node_cnt++;
98 }

```

96 | };

3.9 Sparse Table [31f22a]

```

1 struct SparseTable{
2     vector<vector<int>> st;
3     void build(vector<int> v){
4         int h = __lg(v.size());
5         st.resize(h+1);
6         st[0] = v;
7
8         for (int i=1 ; i<=h ; i++){
9             int gap = (1<<(i-1));
10            for (int j=0 ; j+gap<st[i-1].size() ; j++){
11                st[i].push_back(min(st[i-1][j], st[i-1][j+gap]));
12            }
13        }
14    }
15
16    // 回傳 [ll, rr) 的最小值
17    int query(int ll, int rr){
18        int h = __lg(rr-ll);
19        return min(st[h][ll], st[h][rr-(1<<h)]);
20    }
21 };

```

3.10 Treap2 [3b0cca]

```

1 // 1-based · 請注意 MAX_N 是否足夠大
2 int root = 0;
3 int lc[MAX_N], rc[MAX_N];
4 int pri[MAX_N], val[MAX_N];
5 int sz[MAX_N], tag[MAX_N], fa[MAX_N], total[MAX_N];
6 // tag 為不包含自己 ( 僅要給子樹 ) 的資訊
7 int nodeCnt = 0;
8 int& new_node(int v){
9     nodeCnt++;
10    val[nodeCnt] = v;
11    total[nodeCnt] = v;
12    sz[nodeCnt] = 1;
13    pri[nodeCnt] = rand();
14    return nodeCnt;
15 }
16
17 void apply(int x, int V){
18     val[x] += V;
19     tag[x] += V;
20     total[x] += V*sz[x];
21 }
22
23 void push(int x){
24     if (tag[x]){
25         if (lc[x]) apply(lc[x], tag[x]);
26         if (rc[x]) apply(rc[x], tag[x]);
27     }
28     tag[x] = 0;
29 }
30
31 int pull(int x){
32     if (x){
33         fa[x] = 0;
34         sz[x] = 1+sz[lc[x]]+sz[rc[x]];
35         total[x] = val[x]+total[lc[x]]+total[rc[x]];
36         if (lc[x]) fa[lc[x]] = x;

```

```

36         if (rc[x]) fa[rc[x]] = x;
37     }
38     return x;
39 }
40
41 int merge(int a, int b){
42     if (!a or !b) return a|b;
43     push(a), push(b);
44
45     if (pri[a]>pri[b]){
46         rc[a] = merge(rc[a], b);
47         return pull(a);
48     }else{
49         lc[b] = merge(a, lc[b]);
50         return pull(b);
51     }
52 }
53
54 // [1, k] [k+1, n]
55 void split(int x, int k, int &a, int &b) {
56     if (!x) return a = b = 0, void();
57     push(x);
58     if (sz[lc[x]] >= k) {
59         split(lc[x], k, a, lc[x]);
60         b = x;
61         pull(a); pull(b);
62     }else{
63         split(rc[x], k - sz[lc[x]] - 1, rc[x], b);
64         a = x;
65         pull(a); pull(b);
66     }
67 }
68
69 // functions
70 // 回傳 x 在 Treap 中的位置
71 int get_pos(int x){
72     vector<int> sta;
73     while (fa[x]){
74         sta.push_back(x);
75         x = fa[x];
76     }
77     while (sta.size()){
78         push(x);
79         x = sta.back();
80         sta.pop_back();
81     }
82     push(x);
83
84     int res = sz[x] - sz[rc[x]];
85     while (fa[x]){
86         if (rc[fa[x]]==x){
87             res += sz[fa[x]]-sz[x];
88         }
89         x = fa[x];
90     }
91     return res;
92 }
93
94 // 1-based <前 [1, l-1] 個元素, [l, r] 個元素, [r+1, n] 個元素>
95 array<int, 3> cut(int x, int l, int r){
96     array<int, 3> ret;
97     split(x, r, ret[1], ret[2]);
98     split(ret[1], l-1, ret[0], ret[1]);
99     return ret;

```

```

100 }
101
102 void print(int x){
103     push(x);
104     if (lc[x]) print(lc[x]);
105     cerr << val[x] << " ";
106     if (rc[x]) print(rc[x]);
107 }

```

3.11 Trie [b6475c]

```

1 struct Trie{
2     struct Data{
3         int nxt[2]={0, 0};
4     };
5
6     int sz=0;
7     vector<Data> arr;
8
9     void init(int n){
10        arr.resize(n);
11    }
12
13    void insert(int n){
14        int now=0;
15        for (int i=N ; i>=0 ; i--){
16            int v=(n>>i)&1;
17            if (!arr[now].nxt[v]){
18                arr[now].nxt[v]=++sz;
19            }
20            now=arr[now].nxt[v];
21        }
22    }
23
24    int query(int n){
25        int now=0, ret=0;
26        for (int i=N ; i>=0 ; i--){
27            int v=(n>>i)&1;
28            if (arr[now].nxt[1-v]){
29                ret+=(1<<i);
30                now=arr[now].nxt[1-v];
31            }else if (arr[now].nxt[v]){
32                now=arr[now].nxt[v];
33            }else{
34                return ret;
35            }
36        }
37        return ret;
38    }
39 } tr;
40

```

4 Dynamic-Programming

4.1 Digit DP [133f00]

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 long long l, r;
5 long long dp[20][10][2][2]; // dp[pos][pre][Limit] = 後 pos
6 // 位 · pos 前一位是 pre · (是/否) 有上界 · (是/否) 有前綴零
7 // 的答案數量

```



```

7 long long memorize_search(string &s, int pos, int pre, bool
  limit, bool lead){
8
9 // 已經被找過了・直接回傳值
10 if (dp[pos][pre][limit][lead]!=-1) return dp[pos][pre][
  limit][lead];
11
12 // 已經搜尋完畢・紀錄答案並回傳
13 if (pos==(int)s.size()){
14     return dp[pos][pre][limit][lead] = 1;
15 }
16
17 // 枚舉目前的位數數字是多少
18 long long ans = 0;
19 for (int now=0 ; now<=(limit ? s[pos]-'0' : 9) ; now++){
20     if (now==pre){
21
22         // 1~9 絕對不能連續出現
23         if (pre!=0) continue;
24
25         // 如果已經不在前綴零的範圍內・0 不能連續出現
26         if (lead==false) continue;
27     }
28
29     ans += memorize_search(s, pos+1, now, limit&(now==(s[
  pos]-'0')), lead&(now==0));
30 }
31
32 // 已經搜尋完畢・紀錄答案並回傳
33 return dp[pos][pre][limit][lead] = ans;
34 }
35
36 // 回傳 [0, n] 有多少數字符合條件
37 long long find_answer(long long n){
38     memset(dp, -1, sizeof(dp));
39     string tmp = to_string(n);
40
41     return memorize_search(tmp, 0, 0, true, true);
42 }
43
44 int main(){
45
46     // input
47     cin >> l >> r;
48
49     // output - 計算 [l, r] 有多少數字任意兩個位數都不相同
50     cout << find_answer(r)-find_answer(l-1) << "\n";
51
52     return 0;
53 }

```

4.2 Integer Partition

$dp[i][x]$ = 要將整數 x 拆成 i 堆的「組合數」

$dp[i+1][x+1] += dp[i][x]$ (創造新的一堆)
 $dp[i][x+i] += dp[i][x]$ (把每一堆都增加1)

4.3 Knapsack On Tree [df69b1]

```

1 // 需要重構、需要增加註解
2 #include <bits/stdc++.h>
3 #define F first
4 #define S second
5 #define all(x) begin(x), end(x)

```

```

6 using namespace std;
7
8 #define chmax(a, b) (a) = (a) < (b) ? (b) : (a)
9 #define chmin(a, b) (a) = (a) < (b) ? (a) : (b)
10
11 #define ll long long
12
13 #define FOR(i, a, b) for (int i = a; i <= b; i++)
14
15 int N, W, cur;
16 vector<int> w, v, sz;
17 vector<vector<int>> adj, dp;
18
19 void dfs(int x) {
20     sz[x] = 1;
21     for (int i : adj[x]) dfs(i), sz[x] += sz[i];
22     cur++;
23     // choose x
24     for (int i=w[x] ; i<=W ; i++){
25         dp[cur][i] = dp[cur - 1][i - w[x]] + v[x];
26     }
27     // not choose x
28     for (int i=0 ; i<=W ; i++){
29         chmax(dp[cur][i], dp[cur - sz[x]][i]);
30     }
31 }
32
33 signed main() {
34     cin >> N >> W;
35     adj.resize(N + 1);
36     w.assign(N + 1, 0);
37     v.assign(N + 1, 0);
38     sz.assign(N + 1, 0);
39     dp.assign(N + 2, vector<int>(W + 1, 0));
40     for (int i=1 ; i<=N ; i++){
41         int p; cin >> p;
42         adj[p].push_back(i);
43     }
44
45     for (int i=1 ; i<=N ; i++) cin >> w[i];
46     for (int i=1 ; i<=N ; i++) cin >> v[i];
47     dfs(0);
48     cout << dp[N + 1][W] << "\n";
49 }

```

4.4 SOS DP [8dfa8b]

```

1 // 總時間複雜度為  $O(n \cdot 2^n)$ 
2 // 計算  $dp[i] = i$  所有 bit mask 子集的和
3 for (int i=0 ; i<n ; i++){
4     for (int mask=0 ; mask<(1<n) ; mask++){
5         if ((mask>>i)&1){
6             dp[mask] += dp[mask^(1<i)];
7         }
8     }
9 }

```

5 Geometry

5.1 misc [3ef351]

```

1 using ld = double;
2
3 // 判斷數值正負：{1:正數,0:零,-1:負數}
4 int sign(long long x) {return (x >= 0) ? ((bool)x) : -1; }

```

```

5 int sign(ld x) {return (abs(x) < 1e-9) ? 0 : (x>0 ? 1 : -1); }

```

5.2 convex hull [bbcaaf]

```

1 int sign(long long x) {return (x >= 0) ? ((bool)x) : -1; }
2
3 template<typename T>
4 struct point {
5     T x, y;
6     point() {}
7     point(const T &x0, const T &y0) : x(x0), y(y0) {}
8
9     point operator-(point b) {return {x-b.x, y-b.y}; }
10    bool operator==(point b) {return x==b.x && y==b.y; }
11    T operator^(point b) {return x * b.y - y * b.x; }
12    friend int ori(point a, point b, point c) {
13        return sign((b-a)^(c-a));
14    }
15 };
16
17 template<typename T>
18 struct polygon {
19     vector<point<T>> v;
20     void make_convex_hull(int simple) {
21         auto cmp = [&](point<T> &p, point<T> &q) {
22             return (p.x == q.x) ? (p.y < q.y) : (p.x < q.x);
23         };
24         simple = (bool)simple;
25         sort(v.begin(), v.end(), cmp);
26         v.resize(unique(v.begin(), v.end()) - v.begin());
27         if (v.size() <= 1) return;
28         vector<point<T>> hull;
29         for (int t = 0; t < 2; ++t){
30             int sz = hull.size();
31             for (auto &i:v) {
32                 while (hull.size() >= sz+2 && ori(hull[hull.
33                     size()-2], hull.back(), i) < simple) {
34                     hull.pop_back();
35                 }
36                 hull.push_back(i);
37             }
38             hull.pop_back();
39             reverse(v.begin(), v.end());
40         }
41         swap(hull, v);
42     }
43 };

```

5.3 SVG writer [5bf313]

```

1 class SVG {
2 void p(string_view s) {o << s; }
3 void p(string_view s, auto v, auto... vs) {
4     auto i = s.find('$');
5     o << s.substr(0, i) << v, p(s.substr(i + 1), vs...);
6 }
7 ofstream o;
8 string c = "red";
9 public:
10 SVG(auto f, auto x1, auto y1, auto x2, auto y2) : o(f) {
11     p("<svg xmlns='http://www.w3.org/2000/svg' "
12       "viewBox='$ $ $ $'>\n"
13       "<style>{*{stroke-width:0.5%};}</style>\n",
14       x1, -y2, x2 - x1, y2 - y1);
15 }

```

```

16 }
17 ~SVG() { p("</svg>\n"); }
18 void color(string nc) { c = nc; }
19 void line(auto x1, auto y1, auto x2, auto y2) {
20     p("<line x1='$' y1='$' x2='$' y2='$' stroke='$' />\n",
21     x1, -y1, x2, -y2, c);
22 }
23 void circle(auto x, auto y, auto r) {
24     p("<circle cx='$' cy='$' r='$' stroke='$' "
25     "fill='$' />\n", x, -y, r, c, "#none");
26 }
27 void text(auto x, auto y, string s, int w = 12) {
28     p("<text x='$' y='$' font-size='$px'>$</text>\n",
29     x, -y, w, s);
30 }
31 };
32 // declare : SVG svg("test.svg", -99, -99, 100, 100);

```

5.4 struct point [f21f07]

```

1 template<typename T>
2 struct point {
3     T x, y;
4     point() {}
5     point(const T &x0, const T &y0) : x(x0), y(y0) {}
6     explicit operator point<ld>() {return point<ld>(x, y); }
7 };

```

5.5 point / basic operation [c415da]

```

1 point operator+(point b) {return {x+b.x, y+b.y}; }
2 point operator-(point b) {return {x-b.x, y-b.y}; }
3 point operator*(T b) {return {x*b, y*b}; }
4 point operator/(T b) {return {x/b, y/b}; }
5 bool operator==(point b) {return x==b.x && y==b.y; }
6
7 T operator*(point b) {return x * b.x + y * b.y; }
8 T operator^(point b) {return x * b.y - y * b.x; }

```

5.6 point / operators [fc43e4]

```

1 // 逆時針極角排序
2 bool side() const{return (y == 0) ? (x > 0) : (y < 0); }
3 bool operator<(const point &b) const {
4     return side() == b.side() ?
5         (x*b.y > b.x*y) : side() < b.side();
6 }
7 friend ostream& operator<<(ostream& os, point p) {
8     return os << "(" << p.x << ", " << p.y << ")";
9 }
10 // 判斷 ab 到 ac 的方向 : {1:逆時鐘,0:重疊,-1:順時鐘}
11 friend int ori(point a, point b, point c) {
12     return sign((b-a)^(c-a));
13 }
14 friend bool btw(point a, point b, point c) {
15     return ori(a, b, c) == 0 && sign((a-c)*(b-c)) <= 0;
16 }

```

5.7 point / banana [09fd7c]

```

1 // 判斷線段 ab, cd 是否相交
2 friend bool banana(point a, point b, point c, point d) {
3     if (btw(a, b, c) || btw(a, b, d)

```

```

4     || btw(c, d, a) || btw(c, d, b)) return true;
5 int u = ori(a, b, c) * ori(a, b, d);
6 int v = ori(c, d, a) * ori(c, d, b);
7 return u < 0 && v < 0;
8 }

```

5.8 point / rayHitSeg [db541a]

```

1 // 判斷 "射線 ab" 與 "線段 cd" 是否相交
2 friend bool rayHitSeg(point a,point b,point c,point d) {
3     if (a == b) return btw(c, d, a); // Special case
4     if (((a - b) ^ (c - d)) == 0) {
5         return btw(a, c, b) || btw(a, d, b) || banana(a, b, c
6         , d);
7     }
8     point u = b - a, v = d - c, s = c - a;
9     return sign(s ^ v) * sign(u ^ v) >= 0 && sign(s ^ u)
10        * sign(u ^ v) >= 0 && abs(s ^ u) <= abs(u ^ v);

```

5.9 point / else [f46547]

```

1 // 旋轉 Arg(b) 的角度 (小心溢位)
2 point rotate(point b){return {x*b.x-y*b.y, x*b.y+y*b.x};}
3 // 回傳極座標角度·值域: [-π, +π]
4 friend ld Arg(point b) {
5     return (b.x != 0 || b.y != 0) ? atan2(b.y, b.x) : 0;
6 }
7 friend T abs2(point b) {return b * b; }

```

5.10 struct line [210dc8]

```

1 template<typename T>
2 struct line {
3     point<T> p1, p2;
4     // ax + by + c = 0
5     T a, b, c; // |a|, |b| ≤ 2C, |c| ≤ 8C²
6     line() {}
7     line(const point<T> &x, const point<T> &y) : p1(x), p2(y){
8         build();
9     }
10    void build() {
11        a = p1.y - p2.y;
12        b = p2.x - p1.x;
13        c = (-a*p1.x)-b*p1.y;
14    }
15 };

```

5.11 line / else [09bcc9]

```

1 // 判斷點和有向直線的關係 : {1:左邊,0:在線上,-1:右邊}
2 int ori(point<T> &p) {
3     return sign((p2-p1) ^ (p-p1));
4 }
5 // 判斷直線斜率是否相同
6 bool parallel(line &l) {
7     return ((p1-p2) ^ (l.p1-l.p2)) == 0;
8 }
9 // 兩直線交點
10 point<ld> line_intersection(line &l) {
11     using P = point<ld>;
12     point<T> u = p2-p1, v = l.p2-l.p1, s = l.p1-p1;
13     return P(p1) + P(u) * ((ld(s^v)) / (u^v));
14 }

```

5.12 struct polygon [171a85]

```

1 template<typename T>
2 struct polygon {
3     vector<point<T>> v;
4     polygon() {}
5     polygon(const vector<point<T>> &u) : v(u) {}
6 };

```

5.13 polygon / make convex hull [2bb3ef]

```

1 // simple 為 true 的時候會回傳任意三點不共線的凸包
2 void make_convex_hull(int simple) {
3     auto cmp = [&](point<T> &p, point<T> &q) {
4         return (p.x == q.x) ? (p.y < q.y) : (p.x < q.x);
5     };
6     simple = (bool)simple;
7     sort(v.begin(), v.end(), cmp);
8     v.resize(unique(v.begin(), v.end()) - v.begin());
9     if (v.size() <= 1) return;
10    vector<point<T>> hull;
11    for (int t = 0; t < 2; ++t){
12        int sz = hull.size();
13        for (auto &i:v) {
14            while (hull.size() >= sz+2 && ori(hull[hull.
15                size()-2], hull.back(), i) < simple) {
16                hull.pop_back();
17            }
18            hull.push_back(i);
19        }
20        hull.pop_back();
21        reverse(v.begin(), v.end());
22    }
23    swap(hull, v);
24 }

```

5.14 polygon / in polygon [f11340]

```

1 // 可以在有 n 個點的簡單多邊形內·用 O(n) 判斷一個點 :
2 // {1 : 在多邊形內, 0 : 在多邊形上, -1 : 在多邊形外}
3 int in_polygon(point<T> a){
4     const T MAX_POS = 1e9 + 5; // [記得修改] 座標的最大值
5     point<T> pre = v.back(), b(MAX_POS, a.y + 1);
6     int cnt = 0;
7
8     for (auto &i:v) {
9         if (btw(pre, i, a)) return 0;
10        if (banana(a, b, pre, i)) cnt++;
11        pre = i;
12    }
13
14    return cnt%2 ? 1 : -1;
15 }

```

5.15 polygon / in convex [e64f1e]

```

1  /// 警告：以下所有凸包專用的函式都只接受逆時針排序且任三點不
2  共線的凸包 ///
3  // 可以在有  $n$  個點的凸包內，用  $O(\log n)$  判斷一個點：
4  // {1：在凸包內，0：在凸包邊上，-1：在凸包外}
5  int in_convex(point<T> p) {
6      int n = v.size();
7      int a = ori(v[0], v[1], p), b = ori(v[0], v[n-1], p);
8      if (a < 0 || b > 0) return -1;
9      if (btw(v[0], v[1], p)) return 0;
10     if (btw(v[0], v[n-1], p)) return 0;
11     int l = 1, r = n - 1, mid;
12     while (l + 1 < r) {
13         mid = (l + r) >> 1;
14         if (ori(v[0], v[mid], p) >= 0) l = mid;
15         else r = mid;
16     }
17     int k = ori(v[l], v[r], p);
18     if (k <= 0) return k;
19     return 1;
20 }

```

5.16 polygon / cycle search [fe2f51]

```

1  /// 凸包專用的環狀二分搜，回傳  $\theta$ -based index
2  int cycle_search(auto &f) {
3      int n = v.size(), l = 0, r = n;
4      if (n == 1) return 0;
5      bool rv = f(1, 0);
6      while (r - l > 1) {
7          int m = (l + r) / 2;
8          if (f(0, m) ? rv : f(m, (m + 1) % n)) r = m;
9          else l = m;
10     }
11     return f(1, r % n) ? 1 : r % n;
12 }

```

5.17 polygon / line cut convex [b6a4c8]

```

1  /// 可以在有  $n$  個點的凸包內，用  $O(\log n)$  判斷一條直線：
2  /// {1：穿過凸包，0：剛好切過凸包，-1：沒碰到凸包}
3  int line_cut_convex(line<T> L) {
4      L.build();
5      point<T> p(L.a, L.b);
6      auto gt = [&](int neg) {
7          auto f = [&](int x, int y) {
8              return sign((v[x] - v[y]) * p) == neg;
9          };
10         return -(v[cycle_search(f)] * p);
11     };
12     T x = gt(1), y = gt(-1);
13     if (L.c < x || y < L.c) return -1;
14     return not (L.c == x || L.c == y);
15 }

```

5.18 polygon / convex line intersect [47e699]

```

1  /// return edge endpoint index
2  /// but if (first == second) => intersect at a vertex
3  vector<pair<int, int>> convex_line_intersect(line<T> L) {
4      auto dis = [&](int p) {
5          return (L.p2 - L.p1) ^ (v[p] - L.p1);

```

```

6      };
7      auto gao = [&](int s) {
8          auto f = [&](int i, int j) {
9              return sign(dis(i) - dis(j)) == s;
10         };
11         return cycle_search(f);
12     };
13     int x = gao(1), y = gao(-1), n = v.size();
14     if (sign(dis(x)) < 0 || sign(dis(y)) > 0) return {};
15     if (sign(dis(x)) == 0 || sign(dis(y)) == 0) {
16         int j = ((sign(dis(x)) == 0 ? x : y) + n - 1) % n;
17         vector<pair<int, int>> ret;
18         for (int i = 0; i < 3; ++i, j = (j + 1) % n)
19             if (sign(dis(j)) == 0) ret.emplace_back(j, j);
20         return ret;
21     }
22     auto g = [&](int l, int r, int s) -> pair<int, int> {
23         while ((l + 1) % n != r) {
24             int m = ((l + r + (l < r ? 0 : n)) / 2) % n;
25             if (sign(dis(m)) == s ? l : r) = m;
26         }
27         return {sign(dis(r)) ? l : r, r};
28     };
29     return {g(x, y, 1), g(y, x, -1)};
30 }

```

5.19 polygon / segment pass convex interior [7a3f22]

```

1  /// 可以在有  $n$  個點的凸包內，用  $O(\log n)$  判斷一個線段：
2  /// {1：線段上存在某一點位於凸包內部（邊上不算），
3  /// 0：線段上存在某一點碰到凸包的邊但線段上任一點均不在凸包
4  內部，
5  /// -1：線段完全在凸包外面}
6  int segment_pass_convex_interior(line<T> L) {
7      if (in_convex(L.p1) == 1 || in_convex(L.p2) == 1) return
8          1;
9      L.build();
10     auto res = convex_line_intersect(L);
11     if (res.empty()) return -1;
12     bool cntp = 0, cntn = 0;
13     for (auto &[i, j] : res) if (banana(v[i], v[j], L.p1, L.p2)) {
14         int now = L.ori(v[(i + 1) % v.size()]);
15         cntp |= (now == 1);
16         cntn |= (now == -1);
17     }
18     return (-1 + cntp + cntn);
19 }

```

5.20 polygon / convex tangent point [a6f66b]

```

1  /// 回傳點過凸包的兩條切線的切點的  $\theta$ -based index (不保證兩條
2  切線的順逆時針關係)
3  pair<int, int> convex_tangent_point(point<T> p) {
4      int n = v.size(), z = -1, edg = -1;
5      auto gt = [&](int neg) {
6          auto check = [&](int x) {
7              if (v[x] == p) z = x;
8              if (btw(v[x], v[(x + 1) % n], p)) edg = x;
9              if (btw(v[(x + n - 1) % n], v[x], p)) edg = (x +
10                  n - 1) % n;
11         };
12         auto f = [&](int x, int y) {
13             check(x); check(y);

```

```

12         return ori(p, v[x], v[y]) == neg;
13     };
14     return cycle_search(f);
15 };
16 int x = gt(1), y = gt(-1);
17 if (z != -1) {
18     return {(z + n - 1) % n, (z + 1) % n};
19 }
20 else if (edg != -1) {
21     return {edg, (edg + 1) % n};
22 }
23 else {
24     return {x, y};
25 }
26 }

```

5.21 polygon / halfplane intersection [102d48]

```

1  friend int halfplane_intersection(vector<line<T>> &s, polygon
2  <T> &P) {
3      auto angle_cmp = [&](line<T> &A, line<T> &B) {
4          point<T> a = A.p2-A.p1, b = B.p2-B.p1;
5          return (a < b);
6      };
7      sort(s.begin(), s.end(), angle_cmp); // 線段左側為該線段
8      半平面
9      int L, R, n = s.size();
10     vector<point<T>> px(n);
11     vector<line<T>> q(n);
12     q[L = R = 0] = s[0];
13     for (int i = 1; i < n; ++i) {
14         while (L < R && s[i].ori(px[R-1]) <= 0) --R;
15         while (L < R && s[i].ori(px[L]) <= 0) ++L;
16         q[++R] = s[i];
17         if (q[R].parallel(q[R-1])) {
18             --R;
19             if (q[R].ori(s[i].p1) > 0) q[R] = s[i];
20         }
21         if (L < R) px[R-1] = q[R-1].line_intersection(q[R]);
22     }
23     while (L < R && q[L].ori(px[R-1]) <= 0) --R;
24     P.v.clear();
25     if (R - L <= 1) return 0;
26     px[R] = q[R].line_intersection(q[L]);
27     for (int i = L; i <= R; ++i) P.v.push_back(px[i]);
28     return R - L + 1;
29 }

```

5.22 polygon / minkowski sum [aba519]

```

1  void minkowski(vector<point<T>> a, vector<point<T>> b) {
2      // a, b: convex polygon's inside vector, CCW order
3      auto cmp = [&](auto &a, auto &b) {
4          return tie(a.y, a.x) < tie(b.y, b.x);
5      };
6      auto reorder = [&](auto &u) {
7          auto it = min_element(u.begin(), u.end(), cmp);
8          rotate(u.begin(), it, u.end());
9      };
10     reorder(a); reorder(b);
11     a.push_back(a[0]); a.push_back(a[1]);
12     b.push_back(b[0]); b.push_back(b[1]);
13     v.clear();
14     for (int i = 0, j = 0; i+2<a.size()||j+2<b.size(); ) {

```

```

15     v.push_back(a[i] + b[j]);
16     auto val = (a[i + 1] - a[i]) ^ (b[j + 1] - b[j]);
17     if (val >= 0) i++;
18     if (val <= 0) j++;
19 }
20 }

```

5.23 struct Cir [87198d]

```

1 struct Cir {
2     point<ld> o; ld r;
3     friend ostream& operator<<(ostream& os, Cir c) {
4         return os << "(" << "x" << "+-" [c.o.x >= 0] << abs(c.o.x)
5             << " )^2 + " << "y" << "+-" [c.o.y >= 0] << abs(c.o.y)
6             << " )^2 = " << c.r * c.r;
7     }
8     bool covers(Cir b) {
9         return sqrt((ld)abs2(o - b.o)) + b.r <= r;
10    };

```

5.24 Cir / Cir intersect [330a1c]

```

1 vector<point<ld>> Cir_intersect(Cir c) {
2     ld d2 = abs2(o - c.o), d = sqrt(d2);
3     if (d < max(r, c.r) - min(r, c.r) || d > r + c.r) return {};
4     auto sqdf = [&](ld x, ld y) { return x*x - y*y; };
5     point<ld> u = (o + c.o) / 2 + (o - c.o) * (sqdf(c.r, r) /
6         (2 * d2));
7     ld A = sqrt(sqdf(r + d, c.r) * sqdf(c.r, d - r));
8     point<ld> v = (c.o - o).rotate({0,1}) * A / (2 * d2);
9     if (sign(v.x) == 0 && sign(v.y) == 0) return {u};
10    return {u - v, u + v};

```

5.25 Cir / point tangent [0067e6]

```

1 auto point_tangent(point<ld> p) {
2     vector<point<ld>> res;
3     ld d_sq = abs2(p - o);
4     if (sign(d_sq - r * r) <= 0) {
5         res.pb(p + (p - o).rotate({0, 1}));
6     } else if (d_sq > r * r) {
7         ld s = d_sq - r * r;
8         point<ld> v = p + (o - p) * s / d_sq;
9         point<ld> u = (o - p).rotate({0, 1}) * sqrt(s) * r /
10            d_sq;
11         res.pb(v + u);
12         res.pb(v - u);
13     }
14     return res;

```

5.26 Cir / circle tangent [a3c7fb]

```

1 // returns nothing when two circles are the same
2 auto circle_tangent(Cir c2, int sign1) {
3     // sign1 = 1 for outer tang, -1 for inter tang
4     using ptld = point<ld>;
5     vector<pair<ptld, ptld>> res;
6     ld d_sq = abs2(o - c2.o);
7     if (sign(d_sq) == 0) return res;
8     ld d = sqrt(d_sq);

```

```

9     ptld v = (c2.o - o) / d;
10    ld c = (r - sign1 * c2.r) / d;
11    if (c * c > 1) return res;
12    ld h = sqrt(max((ld)0.0, 1.0 - c * c));
13    for (int sign2 = 1; sign2 >= -1; sign2 -= 2) {
14        ptld n(v.x * c - sign2 * h * v.y, v.y * c + sign2 * h
15            * v.x);
16        ptld p1 = o + n * r;
17        ptld p2 = c2.o + n * (c2.r * sign1);
18        if (sign(p1.x - p2.x) == 0 && sign(p1.y - p2.y) == 0)
19            p2 = p1 + (c2.o - o).rotate({0, 1});
20        res.push_back({p1, p2});
21    }
22    if (sign1 == -1 && sign(r + c2.r - d) == 0) res.pop_back();
23    if (sign1 == 1 && sign(abs(r - c2.r) - d) == 0) res.
24        pop_back();
25    return res;

```

5.27 Cir / minimum enclosing circle [3e1440]

```

1 point<ld> circenter(point<ld> p0, point<ld> p1, point<ld> p2)
2 {
3     // radius = abs(center)
4     p1 = p1 - p0, p2 = p2 - p0;
5     ld x1 = p1.x, y1 = p1.y, x2 = p2.x, y2 = p2.y;
6     ld m = 2. * (x1 * y2 - y1 * x2);
7     point<ld> center(0, 0);
8     center.x = (x1 * x1 * y2 - x2 * x2 * y1 + y1 * y2 * (y1 -
9         y2)) / m;
10    center.y = (x1 * x2 * (x2 - x1) - y1 * y1 * x2 + x1 * y2
11        * y2) / m;
12    return center + p0;
13 }
14 void min_enclosing(vector<point<ld>> p) {
15     shuffle(p.begin(), p.end(), rngf);
16     r = 0.0;
17     point<ld> cent = p[0];
18     for (int i = 1; i < p.size(); ++i) {
19         if (abs2(cent - p[i]) <= r) continue;
20         cent = p[i], r = 0.0;
21         for (int j = 0; j < i; ++j) {
22             if (abs2(cent - p[j]) <= r) continue;
23             cent = (p[i] + p[j]) / 2, r = abs2(p[j] - cent);
24             for (int k = 0; k < j; ++k) {
25                 if (abs2(cent - p[k]) <= r) continue;
26                 cent = circenter(p[i], p[j], p[k]);
27                 r = abs2(p[k] - cent);
28             }
29         }
30     }
31     o = cent;
32     r = sqrt(r);

```

5.28 Geometry Struct 3D [4a50c9]

```

1 using ld = long double;
2
3 // 判斷數值正負：{1:正數,0:零,-1:負數}
4 int sign(long long x) {return (x >= 0) ? ((bool)x) : -1; }
5 int sign(ld x) {return (abs(x) < 1e-9) ? 0 : (x>0 ? 1 : -1); }

```

```

7 template<typename T>
8 struct pt3 {
9     T x, y, z;
10    pt3(){}
11    pt3(const T &x, const T &y, const T &z):x(x),y(y),z(z){}
12    explicit operator pt3<ld>() {return pt3<ld>(x, y, z); }
13
14    pt3 operator+(pt3 b) {return {x+b.x, y+b.y, z+b.z}; }
15    pt3 operator-(pt3 b) {return {x-b.x, y-b.y, z-b.z}; }
16    pt3 operator*(T b) {return {x * b, y * b, z * b}; }
17    pt3 operator/(T b) {return {x / b, y / b, z / b}; }
18    bool operator==(pt3 b){return x==b.x&&y==b.y&&z==b.z;}
19
20    T operator*(pt3 b) {return x*b.x+y*b.y+z*b.z; }
21    pt3 operator^(pt3 b) {
22        return pt3(y * b.z - z * b.y,
23            z * b.x - x * b.z,
24            x * b.y - y * b.x);
25    }
26    friend T abs2(pt3 b) {return b * b; }
27    friend T len (pt3 b) {return sqrt(abs2(b)); }
28    friend ostream& operator<<(ostream& os, pt3 p) {
29        return os << "(" << p.x << ", " <<
30            p.y << ", " << p.z << ")";
31    }
32 };
33
34 template<typename T>
35 struct face {
36     int a, b, c; // 三角形在 vector 裡面的 index
37     pt3<T> q; // 面積向量 ( 朝外 )
38 };
39
40 // 警告；v 在過程中可能被修改，回傳的 face 以修改後的為準
41 // O(n^2) · 最多只有 2n-4 個面
42 // 當凸包退化時會回傳空的凸包，否則回傳凸包上的每個面
43 template<typename T>
44 vector<face<T>> hull3(vector<pt3<T>> &v) {
45     int n = v.size();
46     if (n < 3) return {};
47     // don't use "==" when you use ld
48     sort(all(v), [&](pt3<T> &p, pt3<T> &q) {
49         return sign(p.x - q.x) ? (p.x < q.x) :
50             (sign(p.y - q.y) ? p.y < q.y : p.z < q.z);
51     });
52     v.resize(unique(v.begin(), v.end()) - v.begin());
53     for (int i = 2; i <= n; ++i) {
54         if (i == n) return {};
55         if (sign(len((v[i] - v[0]) ^ (v[i] - v[0])))) {
56             swap(v[2], v[i]);
57             break;
58         }
59     }
60     pt3<T> tmp_q = (v[1] - v[0]) ^ (v[2] - v[0]);
61     for (int i = 3; i <= n; ++i) {
62         if (i == n) return {};
63         if (sign((v[i] - v[0]) * tmp_q)) {
64             swap(v[3], v[i]);
65             break;
66         }
67     }
68
69     vector<face<T>> f;
70     vector<vector<int>> dead(n, vector<int>(n, true));
71     auto add_face = [&](int a, int b, int c) {

```

```

72     f.emplace_back(a, b, c, (v[b] - v[a]) ^ (v[c] - v[a]));
73     dead[a][b] = dead[b][c] = dead[c][a] = false;
74 };
75 add_face(0, 1, 2);
76 add_face(0, 2, 1);
77
78 for (int i = 3; i < n; ++i) {
79     vector<face<T>> f2;
80     for (auto &[a, b, c, q] : f) {
81         if (sign((v[i] - v[a]) * q) > 0)
82             dead[a][b] = dead[b][c] = dead[c][a] = true;
83         else f2.emplace_back(a, b, c, q);
84     }
85     f.clear();
86     for (face<T> &F : f2) {
87         int arr[3] = {F.a, F.b, F.c};
88         for (int j = 0; j < 3; ++j) {
89             int a = arr[j], b = arr[(j + 1) % 3];
90             if (dead[b][a]) add_face(b, a, i);
91         }
92     }
93     f.insert(f.end(), all(f2));
94 }
95 return f;
96 } // 15ef50

```

5.29 Pick's Theorem

給定頂點坐標均是整點的簡單多邊形，面積 = 內部格點數 + 邊上格點數/2 - 1

6 Graph

6.1 SAT [5a6317]

```

1 struct TWO_SAT {
2     int n, N;
3     vector<vector<int>> G, rev_G;
4     deque<bool> used;
5     vector<int> order, comp;
6     deque<bool> assignment;
7     void init(int _n) {
8         n = _n;
9         N = _n * 2;
10        G.resize(N + 5);
11        rev_G.resize(N + 5);
12    }
13    void dfs1(int v) {
14        used[v] = true;
15        for (int u : G[v]) {
16            if (!used[u])
17                dfs1(u);
18        }
19        order.push_back(v);
20    }
21    void dfs2(int v, int c1) {
22        comp[v] = c1;
23        for (int u : rev_G[v]) {
24            if (comp[u] == -1)
25                dfs2(u, c1);
26        }
27    }
28    bool solve() {
29        order.clear();
30        used.assign(N, false);
31        for (int i = 0; i < N; ++i) {

```

```

32         if (!used[i])
33             dfs1(i);
34     }
35     comp.assign(N, -1);
36     for (int i = 0, j = 0; i < N; ++i) {
37         int v = order[N - i - 1];
38         if (comp[v] == -1)
39             dfs2(v, j++);
40     }
41     assignment.assign(n, false);
42     for (int i = 0; i < N; i += 2) {
43         if (comp[i] == comp[i + 1])
44             return false;
45         assignment[i / 2] = (comp[i] > comp[i + 1]);
46     }
47     return true;
48 }
49 // A or B 都是 0-based
50 void add_disjunction(int a, bool na, int b, bool nb) {
51     // na is true => ~a, na is false => a
52     // nb is true => ~b, nb is false => b
53     a = 2 * a ^ na;
54     b = 2 * b ^ nb;
55     int neg_a = a ^ 1;
56     int neg_b = b ^ 1;
57     G[neg_a].push_back(b);
58     G[neg_b].push_back(a);
59     rev_G[b].push_back(neg_a);
60     rev_G[a].push_back(neg_b);
61     return;
62 }
63 void get_result(vector<int>& res) {
64     res.clear();
65     for (int i = 0; i < n; i++)
66         res.push_back(assignment[i]);
67 }
68 };

```

6.2 Augment Path [f8a5dd]

```

1 struct AugmentPath{
2     int n, m;
3     vector<vector<int>> G;
4     vector<int> mx, my;
5     vector<int> visx, visy;
6     int stamp;
7
8     AugmentPath(int _n, int _m) : n(_n), m(_m), G(n), mx(n,
9         -1), my(m, -1), visx(n), visy(m){
10         stamp = 0;
11     }
12
13     void add(int x, int y){
14         G[x].push_back(y);
15     }
16
17     // bb03e2
18     bool dfs1(int now){
19         visx[now] = stamp;
20
21         for (auto x : G[now]){
22             if (my[x]==-1){
23                 mx[now] = x;
24                 my[x] = now;
25                 return true;

```

```

26         }
27     }
28     for (auto x : G[now]){
29         if (visx[my[x]]!=stamp && dfs1(my[x])){
30             mx[now] = x;
31             my[x] = now;
32             return true;
33         }
34     }
35     return false;
36 }
37
38 vector<pair<int, int>> find_max_matching(){
39     vector<pair<int, int>> ret;
40
41     while (true){
42         stamp++;
43         int tmp = 0;
44         for (int i=0; i<n; i++){
45             if (mx[i]==-1 && dfs1(i)) tmp++;
46         }
47         if (tmp==0) break;
48     }
49
50     for (int i=0; i<n; i++){
51         if (mx[i]!=-1){
52             ret.push_back({i, mx[i]});
53         }
54     }
55     return ret;
56 }
57
58 // 645577
59 void dfs2(int now){
60     visx[now] = true;
61
62     for (auto x : G[now]){
63         if (my[x]!=-1 && visy[x]==false){
64             visy[x] = true;
65             dfs2(my[x]);
66         }
67     }
68 }
69
70 // 要先執行 find_max_matching 一次
71 vector<pair<int, int>> find_min_vertex_cover(){
72     fill(visx.begin(), visx.end(), false);
73     fill(visy.begin(), visy.end(), false);
74
75     vector<pair<int, int>> ret;
76     for (int i=0; i<n; i++){
77         if (mx[i]==-1) dfs2(i);
78     }
79
80     for (int i=0; i<n; i++){
81         if (visx[i]==false) ret.push_back({1, i});
82     }
83
84     for (int i=0; i<m; i++){
85         if (visy[i]==true) ret.push_back({2, i});
86     }
87
88     return ret;
89 }

```


6.3 C3C4 [875ea3]

```

1 // 0-based
2 void C3C4(vector<int> deg, vector<array<int, 2>> edges){
3     int N = deg.size();
4     int M = edges.size();
5
6     vector<int> ord(N), rk(N);
7     iota(ord.begin(), ord.end(), 0);
8     sort(ord.begin(), ord.end(), [&](int x, int y) { return
9         deg[x] > deg[y]; });
10    for (int i=0; i<N; i++) rk[ord[i]] = i;
11
12    vector<vector<int>> D(N), adj(N);
13    for (auto [u, v] : edges) {
14        if (rk[u] > rk[v]) swap(u, v);
15        D[u].emplace_back(v);
16        adj[u].emplace_back(v);
17        adj[v].emplace_back(u);
18    }
19
20    vector<int> vis(N);
21
22    int c3 = 0, c4 = 0;
23    for (int x : ord) { // c3
24        for (int y : D[x]) vis[y] = 1;
25        for (int y : D[x]) for (int z : D[y]){
26            c3 += vis[z]; // xyz is C3
27        }
28        for (int y : D[x]) vis[y] = 0;
29    }
30    for (int x : ord) { // c4
31        for (int y : D[x]) for (int z : adj[y])
32            if (rk[z] > rk[x]) c4 += vis[z]++;
33        for (int y : D[x]) for (int z : adj[y])
34            if (rk[z] > rk[x]) --vis[z];
35    } // both are O(M*sqrt(M)), test @ 2022 CCPC guangzhou
36    cout << c4 << "\n";
37 }

```

6.4 Dinic [961b34]

```

1 // 一般圖: O(EV^2)
2 // 二分圖: O(EV)
3 struct Flow{
4     using T = int; // 可以換成別的类型
5     struct Edge{
6         int v; T rc; int rid;
7     };
8     vector<vector<Edge>> G;
9     void add(int u, int v, T c){
10         G[u].push_back({v, c, G[v].size()});
11         G[v].push_back({u, 0, G[u].size()-1});
12     }
13     vector<int> dis, it;
14
15     Flow(int n){
16         G.resize(n);
17         dis.resize(n);
18         it.resize(n);
19     }
20
21     // ce56d6
22     T dfs(int u, int t, T f){
23         if (u == t || f == 0) return f;

```

```

24         for (int &i=it[u]; i<G[u].size(); i++){
25             auto &[v, rc, rid] = G[u][i];
26             if (dis[v]!=dis[u]+1) continue;
27             T df = dfs(v, t, min(f, rc));
28             if (df <= 0) continue;
29             rc -= df;
30             G[v][rid].rc += df;
31             return df;
32         }
33         return 0;
34     }
35
36     // e22e39
37     T flow(int s, int t){
38         T ans = 0;
39         while (true){
40             fill(dis.begin(), dis.end(), INF);
41             queue<int> q;
42             q.push(s);
43             dis[s] = 0;
44
45             while (q.size()){
46                 int u = q.front(); q.pop();
47                 for (auto [v, rc, rid] : G[u]){
48                     if (rc <= 0 || dis[v] < INF) continue;
49                     dis[v] = dis[u] + 1;
50                     q.push(v);
51                 }
52             }
53             if (dis[t]==INF) break;
54
55             fill(it.begin(), it.end(), 0);
56             while (true){
57                 T df = dfs(s, t, INF);
58                 if (df <= 0) break;
59                 ans += df;
60             }
61         }
62         return ans;
63     }
64
65     // the code below constructs minimum cut
66     void dfs_mincut(int now, vector<bool> &vis){
67         vis[now] = true;
68         for (auto &[v, rc, rid] : G[now]){
69             if (vis[v] == false && rc > 0){
70                 dfs_mincut(v, vis);
71             }
72         }
73     }
74
75     vector<pair<int, int>> construct(int n, int s, vector<
76         pair<int, int>> &E){
77         // E is G without capacity
78         vector<bool> vis(n);
79         dfs_mincut(s, vis);
80         vector<pair<int, int>> ret;
81         for (auto &[u, v] : E){
82             if (vis[u] == true && vis[v] == false){
83                 ret.emplace_back(u, v);
84             }
85         }
86         return ret;
87     }

```

6.5 Dominator Tree [52b249]

```

1 // 全部都是 0-based
2 // G 要是無向無權圖
3 // 一開始要初始化 G(N, root)
4 // 用完之後要 build
5 // G[i] = 從 root 走到 i 時，一定要走到的點且離 i 最近
6 struct DominatorTree{
7     int N;
8     vector<vector<int>> G;
9     vector<vector<int>> buckets, rg;
10    // dfn[x] = the DFS order of x
11    // rev[x] = the vertex with DFS order x
12    // par[x] = the parent of x
13    vector<int> dfn, rev, par;
14    vector<int> sdom, dom, idom;
15    vector<int> fa, val;
16    int stamp;
17    int root;
18
19    int operator [] (int x){
20        return idom[x];
21    }
22
23    DominatorTree(int _N, int _root) :
24        N(_N),
25        G(N), buckets(N), rg(N),
26        dfn(N, -1), rev(N, -1), par(N, -1),
27        sdom(N, -1), dom(N, -1), idom(N, -1),
28        fa(N, -1), val(N, -1)
29    {
30        stamp = 0;
31        root = _root;
32    }
33
34    void add_edge(int u, int v){
35        G[u].push_back(v);
36    }
37
38    void dfs(int x){
39        rev[dfn[x] = stamp] = x;
40        fa[stamp] = sdom[stamp] = val[stamp] = stamp;
41        stamp++;
42
43        for (int u : G[x]){
44            if (dfn[u]==-1){
45                dfs(u);
46                par[dfn[u]] = dfn[x];
47            }
48            rg[dfn[u]].push_back(dfn[x]);
49        }
50    }
51
52    int eval(int x, bool first){
53        if (fa[x]==x) return !first ? -1 : x;
54        int p = eval(fa[x], false);
55
56        if (p==-1) return x;
57        if (sdom[val[x]]>sdom[val[fa[x]]]) val[x] = val[fa[x]];
58        fa[x] = p;
59
60        return !first ? p : val[x];
61    }
62 }

```



```

63 void link(int x, int y){
64     fa[x] = y;
65 }
66
67 void build(){
68     dfs(root);
69
70     for (int x=stamp-1 ; x>=0 ; x--){
71         for (int y : rg[x]){
72             sdom[x] = min(sdom[x], sdom[eval(y, true)]);
73         }
74         if (x>0) buckets[sdom[x]].push_back(x);
75         for (int u : buckets[x]){
76             int p = eval(u, true);
77             if (sdom[p]==x) dom[u] = x;
78             else dom[u] = p;
79         }
80         if (x>0) link(x, par[x]);
81     }
82
83     idom[root] = root;
84     for (int x=1 ; x<stamp ; x++){
85         if (sdom[x]!=dom[x]) dom[x] = dom[dom[x]];
86     }
87     for (int i=1 ; i<stamp ; i++) idom[rev[i]] = rev[dom[
88         i]];
89 };

```

6.6 EdgeBCC [d09eb1]

```

1 // d09eb1
2 // 0-based · 支援重邊
3 struct EdgeBCC{
4     int n, m, dep, sz;
5     vector<vector<pair<int, int>>> G;
6     vector<vector<int>> bcc;
7     vector<int> dfn, low, stk, isBridge, bccId;
8     vector<pair<int, int>> edge, bridge;
9
10    EdgeBCC(int _n) : n(_n), m(0), sz(0), dfn(n), low(n), G(n
11        ), bcc(n), bccId(n) {}
12
13    void add_edge(int u, int v) {
14        edge.push_back({u, v});
15        G[u].push_back({v, m});
16        G[v].push_back({u, m++});
17    }
18
19    void dfs(int now, int pre) {
20        dfn[now] = low[now] = ++dep;
21        stk.push_back(now);
22
23        for (auto [x, id] : G[now]){
24            if (!dfn[x]){
25                dfs(x, id);
26                low[now] = min(low[now], low[x]);
27            }else if (id!=pre){
28                low[now] = min(low[now], dfn[x]);
29            }
30        }
31
32        if (low[now]==dfn[now]){
33            if (pre!=-1) isBridge[pre] = true;
34            int u;

```

```

34 do{
35     u = stk.back();
36     stk.pop_back();
37     bcc[sz].push_back(u);
38     bccId[u] = sz;
39     } while (u!=now);
40     sz++;
41 }
42
43 void get_bcc() {
44     isBridge.assign(m, 0);
45     dep = 0;
46     for (int i=0 ; i<n ; i++){
47         if (!dfn[i]) dfs(i, -1);
48     }
49
50     for (int i=0 ; i<m ; i++){
51         if (isBridge[i]){
52             bridge.push_back({edge[i].first , edge[i].
53                 second});
54         }
55     }
56 }
57 };

```

6.7 EnumeratePlanarFace [5498a7]

```

1 // 0-based
2 struct PlanarGraph{
3     int n, m, id;
4     vector<point<int>> v;
5     vector<vector<pair<int, int>>> G;
6     vector<int> conv, nxt, vis;
7
8     PlanarGraph(int n, int m, vector<point<int>> _v) :
9         n(n), m(m), id(0),
10         v(_v), G(n),
11         conv(2*m), nxt(2*m), vis(2*m) {}
12
13    void add_edge(int x, int y){
14        G[x].push_back({y, 2*id});
15        G[y].push_back({x, 2*id+1});
16        conv[2*id] = x;
17        conv[2*id+1] = y;
18        id++;
19    }
20
21    vector<int> enumerate_face(){
22        for (int i=0 ; i<n ; i++){
23            sort(G[i].begin(), G[i].end(), [&](pair<int, int>
24                a, pair<int, int> b){
25                return (v[a.first]-v[i])<(v[b.first]-v[i]);
26            });
27
28            int sz = G[i].size(), pre = sz-1;
29            for (int j=0 ; j<sz ; j++){
30                nxt[G[i][j].second] = G[i][j].second^1;
31                pre = j;
32            }
33        }
34
35        vector<int> ret;
36        for (int i=0 ; i<2*m ; i++){
37            if (vis[i]==false){

```

```

37 int area = 0, now = i;
38 vector<int> pt;
39
40 while (!vis[now]){
41     vis[now] = true;
42     pt.push_back(conv[now]);
43     now = nxt[now];
44 }
45
46 pt.push_back(pt.front());
47 for (int j=0 ; j+1<pt.size() ; j++){
48     area -= (v[pt[j]]^v[pt[j+1]]);
49 }
50
51 // pt = face boundary
52 if (area>0){
53     ret.push_back(area);
54 }else{
55     // pt is outer face
56 }
57 }
58
59 return ret;
60 }
61 };

```

6.8 HLD [f57ec6]

```

1 #include <bits/stdc++.h>
2 #define int long long
3 using namespace std;
4
5 const int N = 100005;
6 vector<int> G[N];
7 struct HLD {
8     vector<int> pa, sz, depth, mxson, topf, id;
9     int n, idcnt = 0;
10    HLD(int _n) : n(_n), pa(_n + 1), sz(_n + 1), depth(_n +
11        1), mxson(_n + 1), topf(_n + 1), id(_n + 1) {}
12    void dfs1(int v = 1, int p = -1) {
13        pa[v] = p; sz[v] = 1; mxson[v] = 0;
14        depth[v] = (p == -1 ? 0 : depth[p] + 1);
15        for (int u : G[v]) {
16            if (u == p) continue;
17            dfs1(u, v);
18            sz[v] += sz[u];
19            if (sz[u] > sz[mxson[v]]) mxson[v] = u;
20        }
21    }
22    void dfs2(int v = 1, int top = 1) {
23        id[v] = ++idcnt;
24        topf[v] = top;
25        if (mxson[v]) dfs2(mxson[v], top);
26        for (int u : G[v]) {
27            if (u == mxson[v] || u == pa[v]) continue;
28            dfs2(u, u);
29        }
30    }
31    // query 為區間資料結構
32    int path_query(int a, int b) {
33        int res = 0;
34        while (topf[a] != topf[b]) { /// 若不在同一條鍊上
35            if (depth[topf[a]] < depth[topf[b]]) swap(a, b);
36            res = max(res, 0ll); // query : l = id[topf[a]],
37            r = id[a]

```

```

36     a = pa[topf[a]];
37 }
38 /// 此時已在同一條鍊上
39 if (depth[a] < depth[b]) swap(a, b);
40 res = max(res, 011); // query : L = id[b], r = id[a]
41 return res;
42 }
43 };

```

6.9 Kosaraju SCC [c7d5aa]

```

1 // 給定一個有向圖・迴傳傳縮點後的圖、SCC 的資訊
2 // 所有點都以 based-0 編號
3 // 函式：
4 // .compress:  $O(n \log n)$  計算  $G_3$ 、SCC、SCC_id 的資訊・並把縮
5 // 點後的結果存在 result 裡
6 // SCC[i] = 某個 SCC 中的所有點
7 // SCC_id[i] = 第 i 個點在第幾個 SCC
8 struct SCC_compress{
9     int N, M, sz;
10    vector<vector<int>> G, inv_G, result;
11    vector<pair<int, int>> edges;
12    vector<bool> vis;
13    vector<int> order;
14
15    vector<vector<int>> SCC;
16    vector<int> SCC_id;
17
18    SCC_compress(int N) :
19        N(N), M(0), sz(0),
20        G(N), inv_G(N),
21        vis(N), SCC_id(N)
22    {}
23
24    vector<int> operator [] (int x){
25        return result[x];
26    }
27
28    void add_edge(int u, int v){
29        G[u].push_back(v);
30        inv_G[v].push_back(u);
31        edges.push_back({u, v});
32        M++;
33    }
34
35    void dfs1(vector<vector<int>> &G, int now){
36        vis[now] = 1;
37        for (auto x : G[now]) if (!vis[x]) dfs1(G, x);
38        order.push_back(now);
39    }
40
41    void dfs2(vector<vector<int>> &G, int now){
42        SCC_id[now] = SCC.size()-1;
43        SCC.back().push_back(now);
44        vis[now] = 1;
45        for (auto x : G[now]) if (!vis[x]) dfs2(G, x);
46    }
47
48    void compress(){
49        fill(vis.begin(), vis.end(), 0);
50        for (int i=0 ; i<N ; i++) if (!vis[i]) dfs1(G, i);
51
52        fill(vis.begin(), vis.end(), 0);
53        reverse(order.begin(), order.end());

```

```

53     for (int i=0 ; i<N ; i++){
54         if (!vis[order[i]]){
55             SCC.push_back(vector<int>());
56             dfs2(inv_G, order[i]);
57         }
58     }
59
60     result.resize(SCC.size());
61     sz = SCC.size();
62     for (auto [u, v] : edges){
63         if (SCC_id[u]!=SCC_id[v]) result[SCC_id[u]].
64             push_back(SCC_id[v]);
65     }
66     for (int i=0 ; i<SCC.size() ; i++){
67         sort(result[i].begin(), result[i].end());
68         result[i].resize(unique(result[i].begin(), result
69             [i].end())-result[i].begin());
70     }
71 }
72 };

```

6.10 Kuhn Munkres [e66c35]

```

1 //  $O(n^3)$  找到最大權匹配
2 struct KuhnMunkres{
3     int n; // max(n, m)
4     vector<vector<int>> G;
5     vector<int> match, lx, ly, visx, visy;
6     vector<int> slack;
7     int stamp = 0;
8
9     KuhnMunkres(int n) : n(n), G(n, vector<int>(n)), lx(n),
10        ly(n), slack(n), match(n), visx(n), visy(n) {}
11
12     void add(int x, int y, int w){
13         G[x][y] = max(G[x][y], w);
14     }
15
16     bool dfs(int i, bool aug){ // aug = true 表示要更新 match
17         if (visx[i]==stamp) return false;
18         visx[i] = stamp;
19
20         for (int j=0 ; j<n ; j++){
21             if (visy[j]==stamp) continue;
22             int d = lx[i]+ly[j]-G[i][j];
23
24             if (d==0){
25                 visy[j] = stamp;
26                 if (match[j]==-1 || dfs(match[j], aug)){
27                     if (aug){
28                         match[j] = i;
29                     }
30                     return true;
31                 }
32             }
33             else{
34                 slack[j] = min(slack[j], d);
35             }
36         }
37         return false;
38     }
39
40     bool augment(){
41         for (int j=0 ; j<n ; j++){
42             if (visy[j]!=stamp && slack[j]==0){
43                 visy[j] = stamp;

```

```

42         if (match[j]==-1 || dfs(match[j], false)){
43             return true;
44         }
45     }
46     return false;
47 }
48
49 void relabel(){
50     int delta = INF;
51     for (int j=0 ; j<n ; j++){
52         if (visy[j]!=stamp) delta = min(delta, slack[j]);
53     }
54     for (int i=0 ; i<n ; i++){
55         if (visx[i]==stamp) lx[i] -= delta;
56     }
57     for (int j=0 ; j<n ; j++){
58         if (visy[j]==stamp) ly[j] += delta;
59         else slack[j] -= delta;
60     }
61 }
62
63 int solve(){
64     for (int i=0 ; i<n ; i++){
65         lx[i] = 0;
66         for (int j=0 ; j<n ; j++){
67             lx[i] = max(lx[i], G[i][j]);
68         }
69     }
70
71     fill(ly.begin(), ly.end(), 0);
72     fill(match.begin(), match.end(), -1);
73
74     for(int i = 0; i < n; i++) {
75         fill(slack.begin(), slack.end(), INF);
76         stamp++;
77         if(dfs(i, true)) continue;
78
79         while(augment()==false) relabel();
80         stamp++;
81         dfs(i, true);
82     }
83
84     int ans = 0;
85     for (int j=0 ; j<n ; j++){
86         if (match[j]!=-1){
87             ans += G[match[j]][j];
88         }
89     }
90     return ans;
91 }
92
93 };
94 };

```

6.11 LCA [5b6a5b]

```

1 // 1-based・可以支援森林・0 是超級源點・所有樹都要跟他建邊
2 struct Tree{
3     int N, M = 0, H;
4     vector<int> parent, dep;
5     vector<vector<int>> G, LCA;
6
7     Tree(int N) : N(N+1), H(__lg(N)+1), parent(N, -1), dep(
8         N), G(N){
9         LCA.resize(H, vector<int>(N, 0));

```

```

9   }
10
11   void add_edge(int u, int v){
12       M++;
13       G[u].push_back(v);
14       G[v].push_back(u);
15   }
16
17   void dfs(int now = 0, int pre = 0){
18       dep[now] = dep[pre]+1;
19       parent[now] = pre;
20       for (auto x : G[now]){
21           if (x==pre) continue;
22           dfs(x, now);
23       }
24   }
25
26   void build_LCA(int root = 0){
27       dfs();
28       for (int i=0 ; i<N ; i++) LCA[0][i] = parent[i];
29       for (int i=1 ; i<H ; i++){
30           for (int j=0 ; j<N ; j++){
31               LCA[i][j] = LCA[i-1][LCA[i-1][j]];
32           }
33       }
34   }
35
36   int jump(int u, int step){
37       for (int i=0 ; i<H ; i++){
38           if (step&(1<<i)) u = LCA[i][u];
39       }
40       return u;
41   }
42
43   int get_LCA(int u, int v){
44       if (dep[u]<dep[v]) swap(u, v);
45       u = jump(u, dep[u]-dep[v]);
46       if (u==v) return u;
47       for (int i=H-1 ; i>=0 ; i--){
48           if (LCA[i][u]!=LCA[i][v]){
49               u = LCA[i][u];
50               v = LCA[i][v];
51           }
52       }
53       return parent[u];
54   }
55 };

```

6.12 MCMF [0d5244]

```

1  struct Flow {
2      struct Edge {
3          int u, rc, k, rv;
4      };
5
6      vector<vector<Edge>> G;
7      vector<int> par, par_eid;
8      Flow(int n) : G(n+1), par(n+1), par_eid(n+1) {}
9
10     // v->u, capacity: c, cost: k
11     void add(int v, int u, int c, int k){
12         G[v].push_back({u, c, k, G[u].size()});
13         G[u].push_back({v, 0, -k, G[v].size()-1});
14     }
15 }

```

```

16 // 6d1140
17 int spfa(int s, int t){
18     fill(par.begin(), par.end(), -1);
19     vector<int> dis(par.size(), INF);
20     vector<bool> in_q(par.size(), false);
21     queue<int> Q;
22     dis[s] = 0;
23     in_q[s] = true;
24     Q.push(s);
25
26     while (!Q.empty()){
27         int v = Q.front();
28         Q.pop();
29         in_q[v] = false;
30
31         for (int i=0 ; i<G[v].size() ; i++){
32             auto [u, rc, k, rv] = G[v][i];
33             if (rc>0 && dis[v]+k<dis[u]){
34                 dis[u] = dis[v]+k;
35                 par[u] = v;
36                 par_eid[u] = i;
37                 if (!in_q[u]) Q.push(u);
38                 in_q[u] = true;
39             }
40         }
41     }
42
43     return dis[t];
44 }
45
46 // return <max flow, min cost>, d7e7ad
47 pair<int, int> flow(int s, int t){
48     int fl = 0, cost = 0, d;
49     while ((d = spfa(s, t))<INF){
50         int cur = INF;
51         for (int v=t ; v!=s ; v=par[v]){
52             cur = min(cur, G[par[v]][par_eid[v]].rc);
53             fl += cur;
54             cost += d*cur;
55         }
56         for (int v=t ; v!=s ; v=par[v]){
57             G[par[v]][par_eid[v]].rc -= cur;
58             G[v][G[par[v]][par_eid[v]].rv].rc += cur;
59         }
60     }
61     return {fl, cost};
62 };

```

6.13 Tarjan SCC [8b2350]

```

1  struct tarjan_SCC {
2      int now_T, now_SCCs;
3      vector<int> dfn, low, SCC;
4      stack<int> S;
5      vector<vector<int>> E;
6      vector<bool> vis, in_stack;
7
8      tarjan_SCC(int n) {
9          init(n);
10     }
11     void init(int n) {
12         now_T = now_SCCs = 0;
13         dfn = low = SCC = vector<int>(n);
14         E = vector<vector<int>>(n);
15         S = stack<int>();

```

```

16         vis = in_stack = vector<bool>(n);
17     }
18     void add(int u, int v) {
19         E[u].push_back(v);
20     }
21     void build() {
22         for (int i = 0; i < dfn.size(); ++i) {
23             if (!dfn[i]) dfs(i);
24         }
25     }
26     void dfs(int v) {
27         now_T++;
28         vis[v] = in_stack[v] = true;
29         dfn[v] = low[v] = now_T;
30         S.push(v);
31         for (auto &i:E[v]) {
32             if (!vis[i]) {
33                 vis[i] = true;
34                 dfs(i);
35                 low[v] = min(low[v], low[i]);
36             }
37             else if (in_stack[i]) {
38                 low[v] = min(low[v], dfn[i]);
39             }
40         }
41         if (low[v] == dfn[v]) {
42             int tmp;
43             do {
44                 tmp = S.top();
45                 S.pop();
46                 SCC[tmp] = now_SCCs;
47                 in_stack[tmp] = false;
48             } while (tmp != v);
49             now_SCCs += 1;
50         }
51     }
52 };

```

6.14 Theorem

- 任意圖
 - 最大匹配 + 最小邊覆蓋 = n (不能有孤點)
 - 點覆蓋的補集是獨立集。最小點覆蓋 + 最大獨立集 = n
 - $w(\text{最小權點覆蓋}) + w(\text{最大權獨立集}) = \sum w_v$
 - (帶點權的二分圖可以用最小割解。構造請參考 Augment Path.cpp)
- 二分圖
 - 最小點覆蓋 = 最大匹配 = n - 最大獨立集
- 只有帶權的二分圖
 - w-vertex-cover (帶權點覆蓋): 每條邊的兩個連接點被選中的次數總和至少要是 w_e 。
 - w-weight matching (帶權匹配)
 - minimum vertex count of w-vertex-cover = maximum weight count of w-weight matching (一個點可以被選很多次。但邊不行)
- 點、邊都帶權的二分圖的定理
 - b-matching: 假設 v 的點權是 b_v 。那所有 v 的匹配邊 e 的權重都要滿足 $\sum w_e \leq b_v$ 。
 - The maximum w-weight of a b-matching equals the minimum b-weight of vertices in a w-vertex-cover.

6.15 Tree Isomorphism [cd2bbc]

```

1 #include <bits/stdc++.h>
2 #pragma GCC optimize("O3,unroll-loops")
3 #define fastio ios::sync_with_stdio(0), cin.tie(0), cout.tie
  (0)
4 #define dbg(x) cerr << #x << " = " << x << endl
5 #define int long long
6 using namespace std;
7
8 // declare
9 const int MAX_SIZE = 2e5+5;
10 const int INF = 9e18;
11 const int MOD = 1e9+7;
12 const double EPS = 1e-6;
13 typedef vector<vector<int>> Graph;
14 typedef map<vector<int>, int> Hash;
15
16 int n, a, b;
17 int id1, id2;
18 pair<int, int> c1, c2;
19 vector<int> sz1(MAX_SIZE), sz2(MAX_SIZE);
20 vector<int> we1(MAX_SIZE), we2(MAX_SIZE);
21 Graph g1(MAX_SIZE), g2(MAX_SIZE);
22 Hash m1, m2;
23 int testcase=0;
24
25 void centroid(Graph &g, vector<int> &s, vector<int> &w, pair<
  int, int> &rec, int now, int pre){
26     s[now]=1;
27     w[now]=0;
28     for (auto x : g[now]){
29         if (x!=pre){
30             centroid(g, s, w, rec, x, now);
31             s[now]+=s[x];
32             w[now]=max(w[now], s[x]);
33         }
34     }
35
36     w[now]=max(w[now], n-s[now]);
37     if (w[now]<=n/2){
38         if (rec.first==0) rec.first=now;
39         else rec.second=now;
40     }
41 }
42
43 int dfs(Graph &g, Hash &m, int &id, int now, int pre){
44     vector<int> v;
45     for (auto x : g[now]){
46         if (x!=pre){
47             int add=dfs(g, m, id, x, now);
48             v.push_back(add);
49         }
50     }
51     sort(v.begin(), v.end());
52
53     if (m.find(v)!=m.end()){
54         return m[v];
55     }else{
56         m[v]=++id;
57         return id;
58     }
59 }
60
61 void solve1(){
62

```

```

63
64 // init
65 id1=0;
66 id2=0;
67 c1={0, 0};
68 c2={0, 0};
69 fill(sz1.begin(), sz1.begin()+n+1, 0);
70 fill(sz2.begin(), sz2.begin()+n+1, 0);
71 fill(we1.begin(), we1.begin()+n+1, 0);
72 fill(we2.begin(), we2.begin()+n+1, 0);
73 for (int i=1 ; i<=n ; i++){
74     g1[i].clear();
75     g2[i].clear();
76 }
77 m1.clear();
78 m2.clear();
79
80 // input
81 cin >> n;
82 for (int i=0 ; i<=n-1 ; i++){
83     cin >> a >> b;
84     g1[a].push_back(b);
85     g1[b].push_back(a);
86 }
87 for (int i=0 ; i<=n-1 ; i++){
88     cin >> a >> b;
89     g2[a].push_back(b);
90     g2[b].push_back(a);
91 }
92
93 // get tree centroid
94 centroid(g1, sz1, we1, c1, 1, 0);
95 centroid(g2, sz2, we2, c2, 1, 0);
96
97 // process
98 int res1=0, res2=0, res3=0;
99 if (c2.second!=0){
100     res1=dfs(g1, m1, id1, c1.first, 0);
101     m2=m1;
102     id2=id1;
103     res2=dfs(g2, m1, id1, c2.first, 0);
104     res3=dfs(g2, m2, id2, c2.second, 0);
105 }else if (c1.second!=0){
106     res1=dfs(g2, m1, id1, c2.first, 0);
107     m2=m1;
108     id2=id1;
109     res2=dfs(g1, m1, id1, c1.first, 0);
110     res3=dfs(g1, m2, id2, c1.second, 0);
111 }else{
112     res1=dfs(g1, m1, id1, c1.first, 0);
113     res2=dfs(g2, m1, id1, c2.first, 0);
114 }
115
116 // output
117 cout << (res1==res2 || res1==res3 ? "YES" : "NO") << endl
  ;
118
119 return;
120 }
121
122 signed main(void){
123     fastio;
124
125     int t=1;
126     cin >> t;
127     while (t--){

```

```

128         solve1();
129     }
130     return 0;
131 }

```

6.16 prufer reverse [2f831d]

```

1
2 // K_n 有  $n^{n-2}$  種生成樹
3 vector<pair<int, int>> prufer_reverse(vector<int> &v) {
4     int n = v.size() + 2, mn = 1, mx = n; // 1-based
5     vector<int> deg(mx + 1);
6     for (auto &i:v) deg[i] += 1;
7     int ind = mn;
8     while (deg[ind] != 0) ind += 1;
9     int leaf = ind;
10    vector<pair<int, int>> res;
11    for (auto &i:v) {
12        res.emplace_back(leaf, i);
13        deg[i] -= 1;
14        if (deg[i] == 0 && i < ind) leaf = i;
15        else {
16            do ind += 1; while (deg[ind] != 0);
17            leaf = ind;
18        }
19    }
20    res.emplace_back(leaf, mx);
21    return res;
22 }

```

6.17 圓方樹 [1a1189]

```

1
2 struct BCC_AP { // 0-based, remember to build
3     int n, nbcc; // 注意孤點不會有任何方點連接
4     vector<vector<int>> E, F; // id >= n: 方點
5     vector<int> pa, dep, low, stk, paf, depf;
6     void dfs(int v, int p) {
7         dep[v] = low[v] = ~p ? dep[p] + 1 : 0;
8         stk.push_back(v), pa[v] = p;
9         for (auto& u : E[v]) if (u != p) {
10             if (low[u] == -1) {
11                 dfs(u, v), low[v] = min(low[v], low[u]);
12                 if (low[u] >= dep[v]) {
13                     int id = nbcc++, x;
14                     do {
15                         x = stk.back(), stk.pop_back();
16                         F[id + n].push_back(x), F[x].
17                             push_back(id + n);
18                     } while (x != u);
19                     F[id + n].push_back(v), F[v].push_back(id
20                         + n);
21                 } else low[v] = min(low[v], dep[u]);
22             }
23         }
24         // 判斷原圖上的一個點是不是割點
25         bool is_cut(int x) { return F[x].size() != 1; }
26         // 回傳第 id 個 bcc 的每個點 (0-based)
27         vector<int> bcc(int id) { return F[id + n]; }
28         // 對於邊 (u, v) 回傳包含它的 bcc id (每條邊都恰好被一個
29         // bcc 包含)
30         int bcc_id(int u, int v) { // starts from 0
31             return paf[depf[u] < depf[v] ? v : u] - n; }

```

```

30 // 計算在新圖上的 depf, paf 陣列・呼叫 bcc_id() 的時候會
    用到
31 void dfs2(int v, int p) {
32     depf[v] = ~p ? depf[p] + 1 : 0, paf[v] = p;
33     for (int u : F[v]) if (u != p) dfs2(u, v);
34 }
35 void build() {
36     low.assign(n, -1);
37     for (int i = 0; i < n; ++i) if (low[i] == -1)
38         dfs(i, -1), dfs2(i, -1);
39 }
40 void add_edge(int u, int v) {
41     E[u].push_back(v), E[v].push_back(u);
42 }
43 BCC_AP (int _n) : n(_n), nbcc(0), E(n), F(2 * n), pa(n),
    dep(n), low(n), stk(), paf(n * 2), depf(n * 2) {}
44 };

```

6.18 最大權閉圖 [6ca663]

```

1 // 邊 u → v 表示選 u 就要選 v (0-based)
2 // 保證回傳值非負
3 // 構造：從 S 開始 dfs・不走最小割的邊・
4 // 所有經過的點就是要選的那些點。
5 // 一般圖：O(n^2m) / 二分圖：O(m√n)
6 template<typename U>
7 U maximum_closure(vector<U> w, vector<pair<int,int>> EV) {
8     int n = w.size(), S = n + 1, T = n + 2;
9     Flow G(T + 5); // Graph/Dinic.cpp
10    U sum = 0;
11    for (int i = 0; i < n; ++i) {
12        if (w[i] > 0) {
13            G.add(S, i, w[i]);
14            sum += w[i];
15        }
16        else if (w[i] < 0) {
17            G.add(i, T, abs(w[i]));
18        }
19    }
20    for (auto &[u, v] : EV) { // 請務必確保 INF > Σ|w_i|
21        G.add(u, v, INF);
22    }
23    U cut = G.flow(S, T);
24    return sum - cut;
25 }

```

7 Math

7.1 Burnside's Lemma

$$\sum_{k=1}^n \frac{c(k)}{n}$$

- n : 有多少種置換方式 (例如：旋轉方式)
- $c(k)$: 所有可能中・經過 k 次旋轉後・仍不會和別人相同的方式的數量

7.2 CRT [6a6da6]

```

1 using ll = __int128;
2 //a * p.first + b * p.second = gcd(a, b)
3 pair<ll, ll> extgcd(ll a, ll b) {

```

```

5     if (b == 0) return {1, 0};
6     auto [y, x] = extgcd(b, a % b);
7     return pair<ll, ll>(x, y - (a / b) * x);
8 }
9
10 pair<ll, ll> CRT(ll x1, ll m1, ll x2, ll m2) {
11     ll g = __gcd(m1, m2);
12     if ((x2 - x1) % g) return make_pair(-1, -1); // no sol
13     m1 /= g, m2 /= g;
14     pair<ll, ll> p = extgcd(m1, m2);
15     ll lcm = m1 * m2 * g;
16     ll res = p.first * (x2 - x1) * m1 + x1;
17     // be careful with overflow
18     return make_pair((res % lcm + lcm) % lcm, lcm);
19 }
20
21 pair<ll, ll> CRT_arr(vector<ll> &a, vector<ll> &m) {
22     pair<ll, ll> res(a[0], m[0]);
23     for (int i = 1; i < a.size(); ++i) {
24         res = CRT(res.first, res.second, a[i], m[i]);
25         if (res.first == -1) break;
26     }
27     return res;
28 }

```

7.3 Catalan Number

任意括號序列： $C_n = \frac{1}{n+1} \binom{2n}{n}$

7.4 Floor Sum [5d1c9e]

```

1 // Returns sum_{i=0}^{n-1} floor((a*i + b) / m)
2 // log( max(a, m) )
3 int floor_sum(int n, int m, int a, int b) {
4     int ans = 0;
5     while (true) {
6         if (a >= m) {
7             ans += (n - 1) * n * (a / m) / 2;
8             a %= m;
9         }
10        if (b >= m) {
11            ans += n * (b / m);
12            b %= m;
13        }
14        int y_max = a * n + b;
15        if (y_max < m) break;
16        n = y_max / m;
17        b = y_max % m;
18        swap(m, a);
19    }
20    return ans;
21 }

```

7.5 Josephus Problem [e0ed50]

```

1 // 有 n 個人・第偶數個報數的人被刪掉・問第 k 個被踢掉的是誰
2 int solve(int n, int k){
3     if (n==1) return 1;
4     if (k<=(n+1)/2){
5         if (2*k>n) return 2*k%n;
6         else return 2*k;
7     }else{
8         int res=solve(n/2, k-(n+1)/2);
9         if (n&1) return 2*res+1;
10        else return 2*res-1;

```

```

11    }
12 }

```

7.6 Lagrange any x [1f2c26]

```

1 // init: (x1, y1), (x2, y2) in a vector
2 struct Lagrange{
3     int n;
4     vector<pair<int, int>> v;
5
6     Lagrange(vector<pair<int, int>> &_v){
7         n = _v.size();
8         v = _v;
9     }
10
11    // O(n^2 log MAX_A)
12    int solve(int x){
13        int ret = 0;
14        for (int i=0; i<n; i++){
15            int now = v[i].second;
16            for (int j=0; j<n; j++){
17                if (i==j) continue;
18                now *= ((x-v[j].first+MOD)%MOD);
19                now %= MOD;
20                now *= (qp((v[i].first-v[j].first+MOD)%MOD,
21                    MOD-2)+MOD)%MOD;
22                now %= MOD;
23            }
24            ret = (ret+now)%MOD;
25        }
26        return ret;
27    }
28 };

```

7.7 Lagrange continuous x [57536a]

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 const int MAX_N = 5e5 + 10;
5 const int mod = 1e9 + 7;
6
7 long long inv_fac[MAX_N];
8
9 inline int fp(long long x, int y) {
10    int ret = 1;
11    for (; y >= 1) {
12        ret = (y & 1) ? (ret * x % mod) : ret;
13        x = x * x % mod;
14    }
15    return ret;
16 }
17
18 // TO USE THIS TEMPLATE, YOU MUST MAKE SURE THAT THE MOD
    NUMBER IS A PRIME.
19 struct Lagrange {
20     /*
21        Initialize a polynomial with f(x_0), f(x_0 + 1), ..., f(
        x_0 + n).
22        This determines a polynomial f(x) whose degree is at most
        n.
23        Then you can call sample(x) and you get the value of f(x)
        .
24        Complexity of init() and sample() are both O(n).

```

```

25 */
26 int m, shift; // m = n + 1
27 vector<int> v, mul;
28 // You can use this function if you don't have inv_fac array
   already.
29 void construct_inv_fac() {
30     long long fac = 1;
31     for (int i = 2; i < MAX_N; ++i) {
32         fac = fac * i % mod;
33     }
34     inv_fac[MAX_N - 1] = fp(fac, mod - 2);
35     for (int i = MAX_N - 1; i >= 1; --i) {
36         inv_fac[i - 1] = inv_fac[i] * i % mod;
37     }
38 }
39 // You call init() many times without having a second
   instance of this struct.
40 void init(int X_0, vector<int> &u) {
41     v = u;
42     shift = ((1 - X_0) % mod + mod) % mod;
43     if (v.size() == 1) v.push_back(v[0]);
44     m = v.size();
45     mul.resize(m);
46 }
47 // You can use sample(x) instead of sample(x % mod).
48 int sample(int x) {
49     x = ((long long)x + shift) % mod;
50     x = (x < 0) ? (x + mod) : x;
51     long long now = 1;
52     for (int i = m; i >= 1; --i) {
53         mul[i - 1] = now;
54         now = now * (x - i) % mod;
55     }
56     int ret = 0;
57     bool neg = (m - 1) & 1;
58     now = 1;
59     for (int i = 1; i <= m; ++i) {
60         int up = now * mul[i - 1] % mod;
61         int down = inv_fac[m - i] * inv_fac[i - 1] % mod;
62         int tmp = ((long long)v[i - 1] * up % mod) * down
           % mod;
63         ret += (neg && tmp) ? (mod - tmp) : (tmp);
64         ret = (ret >= mod) ? (ret - mod) : ret;
65         now = now * (x - i) % mod;
66         neg ^= 1;
67     }
68     return ret;
69 }
70 };
71
72 int main() {
73     int n; cin >> n;
74     vector<int> v(n);
75     for (int i = 0; i < n; ++i) {
76         cin >> v[i];
77     }
78     Lagrange L;
79     L.construct_inv_fac();
80     L.init(0, v);
81     int x; cin >> x;
82     cout << L.sample(x);
83 }

```

7.8 Linear Mod Inverse [ecf71e]

```

1 // 線性求 1-based a[i] 對 p 的乘法反元素
2 vector<int> s(n+1, 1), invS(n+1), invA(n+1);
3 for (int i=1; i<=n; i++) s[i] = s[i-1]*a[i]%p;
4 invS[n] = qp(s[n], p-2, p);
5 for (int i=n; i>=1; i--) invS[i-1] = invS[i]*a[i]%p;
6 for (int i=1; i<=n; i++) invA[i] = invS[i]*s[i-1]%p;

```

7.9 Lucas's Theorem [b37dcf]

```

1 // 對於很大的  $C^n_{m}$  對質數  $p$  取模。只要  $p$  不大就可以用。
2 int Lucas(int n, int m, int p){
3     if (m==0) return 1;
4     return (C(n%p, m%p, p)*Lucas(n/p, m/p, p)%p);
5 }

```

7.10 Matrix [8d1a23]

```

1 struct Matrix{
2     int n, m;
3     vector<vector<int>> arr;
4
5     Matrix(int _n, int _m){
6         n = _n;
7         m = _m;
8         arr.assign(n, vector<int>(m));
9     }
10
11     vector<int> & operator [] (int i){
12         return arr[i];
13     }
14
15     Matrix operator * (Matrix b){
16         Matrix ret(n, b.m);
17         for (int i=0; i<n; i++){
18             for (int j=0; j<b.m; j++){
19                 for (int k=0; k<m; k++){
20                     ret.arr[i][j] += arr[i][k]*b.arr[k][j]%
                       MOD;
21                     ret.arr[i][j] %= MOD;
22                 }
23             }
24         }
25         return ret;
26     }
27
28     Matrix pow(int p){
29         Matrix ret(n, n), mul = *this;
30         for (int i=0; i<n; i++){
31             ret.arr[i][i] = 1;
32         }
33
34         for (; p; p>>=1){
35             if (p&1) ret = ret*mul;
36             mul = mul*mul;
37         }
38
39         return ret;
40     }
41
42     int det(){
43         vector<vector<int>> arr = this->arr;
44         bool flag = false;
45         for (int i=0; i<n; i++){
46             int target = -1;

```

```

48         for (int j=i; j<n; j++){
49             if (arr[j][i]){
50                 target = j;
51                 break;
52             }
53         }
54         if (target==-1) return 0;
55         if (i!=target){
56             swap(arr[i], arr[target]);
57             flag = !flag;
58         }
59
60         for (int j=i+1; j<n; j++){
61             if (!arr[j][i]) continue;
62             int freq = arr[j][i]*qp(arr[i][i], MOD-2)%MOD;
63             ;
64             for (int k=i; k<n; k++){
65                 arr[j][k] -= freq*arr[i][k];
66                 arr[j][k] = (arr[j][k]%MOD+MOD)%MOD;
67             }
68         }
69
70         int ret = !flag ? 1 : MOD-1;
71         for (int i=0; i<n; i++){
72             ret *= arr[i][i];
73             ret %= MOD;
74         }
75         return ret;
76     }
77 };

```

7.11 Matrix 01 [8d542a]

```

1 const int MAX_N = (1LL<<12);
2 struct Matrix{
3     int n, m;
4     vector<bitset<MAX_N>> arr;
5
6     Matrix(int _n, int _m){
7         n = _n;
8         m = _m;
9         arr.resize(n);
10    }
11
12    Matrix operator * (Matrix b){
13        Matrix b_t(b.m, b.n);
14        for (int i=0; i<b.n; i++){
15            for (int j=0; j<b.m; j++){
16                b_t.arr[j][i] = b.arr[i][j];
17            }
18        }
19
20        Matrix ret(n, b.m);
21        for (int i=0; i<n; i++){
22            for (int j=0; j<b.m; j++){
23                ret.arr[i][j] = (arr[i]&b_t.arr[j]).count()
                  &1;
24            }
25        }
26        return ret;
27    }
28 };

```


7.12 Matrix Tree Theorem

目標：給定一張無向圖，問他的生成樹數量。
方法：先把所有自環刪掉，定義 Q 為以下矩陣

$$Q_{i,j} = \begin{cases} \deg(v_i) & \text{if } i = j \\ -(邊v_i v_j \text{ 的數量}) & \text{otherwise} \end{cases}$$

接著刪掉 Q 的第一個 row 跟 column，它的 determinant 就是答案。
目標：給定一張有向圖，問他的以 r 為根，可以走到所有點生成樹數量。

方法：先把所有自環刪掉，定義 Q 為以下矩陣

$$Q_{i,j} = \begin{cases} \deg_{in}(v_i) & \text{if } i = j \\ -(邊v_i v_j \text{ 的數量}) & \text{otherwise} \end{cases}$$

接著刪掉 Q 的第 r 個 row 跟 column，它的 determinant 就是答案。

7.13 Miller Rabin [5694bf]

```
1 // O(k log^3 n), k = llsprp.size()
2 using Uint = unsigned long long;
3 Uint modmul(Uint a, Uint b, Uint m) {
4     int ret = a*b - m*(Uint)((long double)a*b/m);
5     return ret + m*(ret < 0) - m*(ret >=(int)m);
6 }
7
8 int qp(int b, int p, int m){
9     int ret = 1;
10    for (; p; p>>=1){
11        if (p&1) ret = modmul(ret, b, m);
12        b = modmul(b, b, m);
13    }
14    return ret;
15 }
16
17 // ed23aa
18 vector<int> llsprp = {2, 325, 9375, 28178, 450775, 9780504,
19     1795265022};
20 bool is_prime(int n, vector<int> sprp = llsprp){
21     if (n==2) return 1;
22     if (n<2 || n%2==0) return 0;
23
24     int t = 0;
25     int u = n-1;
26     for (; u%2==0 ; t++) u>>=1;
27
28     for (int i=0 ; i<sprp.size() ; i++){
29         int a = sprp[i]%n;
30         if (a==0 || a==1 || a==n-1) continue;
31         int x = qp(a, u, n);
32         if (x==1 || x==n-1) continue;
33         for (int j=0 ; j<t ; j++){
34             x = modmul(x, x, n);
35             if (x==1) return 0;
36             if (x==n-1) break;
37         }
38         if (x==n-1) continue;
39         return false;
40     }
41
42     return true;
43 }
```

7.14 Pollard Rho [a5daef]

```
1 mt19937 seed(chrono::steady_clock::now().time_since_epoch().
2     count());
3 int rnd(int l, int r){
4     return uniform_int_distribution<int>(l, r)(seed);
5 }
6 // O(n^(1/4)) 回傳 1 或自己的因數、記得先判斷 n 是不是質數
7 // (用 Miller-Rabin)
8 // c1670c
9 int Pollard_Rho(int n){
10    int s = 0, t = 0;
11    int c = rnd(1, n-1);
12
13    int step = 0, goal = 1;
14    int val = 1;
15
16    for (goal=1 ; ; goal<=1, s=t, val=1){
17        for (step=1 ; step<=goal ; step++){
18            t = ((__int128)t*t+c)%n;
19            val = ((__int128)val*abs(t-s)%n;
20
21            if ((step % 127) == 0){
22                int d = __gcd(val, n);
23                if (d>1) return d;
24            }
25        }
26
27        int d = __gcd(val, n);
28        if (d>1) return d;
29    }
30 }
```

7.15 Polynomial [51ca3b]

```
1 struct Poly {
2     int len, deg;
3     int *a;
4     // Len = 2^k >= the original length
5     Poly(): len(0), deg(0), a(nullptr) {}
6     Poly(int _n) {
7         len = 1;
8         deg = _n - 1;
9         while (len < _n) len <= 1;
10        a = (ll*) calloc(len, sizeof(ll));
11    }
12    Poly(int l, int d, int *b) {
13        len = l;
14        deg = d;
15        a = b;
16    }
17    void resize(int _n) {
18        int len1 = 1;
19        while (len1 < _n) len1 <= 1;
20        int *res = (ll*) calloc(len1, sizeof(ll));
21        for (int i = 0; i < min(len, _n); i++) {
22            res[i] = a[i];
23        }
24        len = len1;
25        deg = _n - 1;
26        free(a);
27        a = res;
28    }
29 }
```

```
29 Poly& operator=(const Poly rhs) {
30     this->len = rhs.len;
31     this->deg = rhs.deg;
32     this->a = (ll*)realloc(this->a, sizeof(ll) * len);
33     copy(rhs.a, rhs.a + len, this->a);
34     return *this;
35 }
36
37 Poly operator*(Poly rhs) {
38     int l1 = this->len, l2 = rhs.len;
39     int d1 = this->deg, d2 = rhs.deg;
40     while (l1 > 0 and this->a[l1 - 1] == 0) l1--;
41     while (l2 > 0 and rhs.a[l2 - 1] == 0) l2--;
42     int l = 1;
43     while (l < max(l1 + l2 - 1, d1 + d2 + 1)) l <= 1;
44     int *x, *y, *res;
45     x = (ll*) calloc(l, sizeof(ll));
46     y = (ll*) calloc(l, sizeof(ll));
47     res = (ll*) calloc(l, sizeof(ll));
48     copy(this->a, this->a + l1, x);
49     copy(rhs.a, rhs.a + l2, y);
50     ntt.tran(l, x); ntt.tran(l, y);
51     FOR (i, 0, l - 1)
52         res[i] = x[i] * y[i] % mod;
53     ntt.tran(l, res, true);
54     free(x); free(y);
55     return Poly(l, d1 + d2, res);
56 }
57
58 Poly operator+(Poly rhs) {
59     int l1 = this->len, l2 = rhs.len;
60     int l = max(l1, l2);
61     Poly res;
62     res.len = l;
63     res.deg = max(this->deg, rhs.deg);
64     res.a = (ll*) calloc(l, sizeof(ll));
65     FOR (i, 0, l1 - 1) {
66         res.a[i] += this->a[i];
67         if (res.a[i] >= mod) res.a[i] -= mod;
68     }
69     FOR (i, 0, l2 - 1) {
70         res.a[i] += rhs.a[i];
71         if (res.a[i] >= mod) res.a[i] -= mod;
72     }
73     return res;
74 }
75
76 Poly operator-(Poly rhs) {
77     int l1 = this->len, l2 = rhs.len;
78     int l = max(l1, l2);
79     Poly res;
80     res.len = l;
81     res.deg = max(this->deg, rhs.deg);
82     res.a = (ll*) calloc(l, sizeof(ll));
83     FOR (i, 0, l1 - 1) {
84         res.a[i] += this->a[i];
85         if (res.a[i] >= mod) res.a[i] -= mod;
86     }
87     FOR (i, 0, l2 - 1) {
88         res.a[i] -= rhs.a[i];
89         if (res.a[i] < 0) res.a[i] += mod;
90     }
91     return res;
92 }
93
94 Poly operator*(const int rhs) {
95     Poly res;
96     res = *this;
97     FOR (i, 0, res.len - 1) {
98         res.a[i] = res.a[i] * rhs % mod;
99     }
100 }
```

```

95     if (res.a[i] < 0) res.a[i] += mod;
96 }
97 return res;
98 }
99 Poly(vector<int> f) {
100     int _n = f.size();
101     len = 1;
102     deg = _n - 1;
103     while (len < _n) len <= 1;
104     a = (ll*) calloc(len, sizeof(ll));
105     FOR (i, 0, deg) a[i] = f[i];
106 }
107 Poly derivative() {
108     Poly g(this->deg);
109     FOR (i, 1, this->deg) {
110         g.a[i - 1] = this->a[i] * i % mod;
111     }
112     return g;
113 }
114 Poly integral() {
115     Poly g(this->deg + 2);
116     FOR (i, 0, this->deg) {
117         g.a[i + 1] = this->a[i] * ::inv(i + 1) % mod;
118     }
119     return g;
120 }
121 Poly inv(int len1 = -1) {
122     if (len1 == -1) len1 = this->len;
123     Poly g(1); g.a[0] = ::inv(a[0]);
124     for (int l = 1; l < len1; l <= 1) {
125         Poly t; t = *this;
126         t.resize(l < 1);
127         t = g * g * t;
128         t.resize(l < 1);
129         Poly g1 = g * 2 - t;
130         swap(g, g1);
131     }
132     return g;
133 }
134 Poly ln(int len1 = -1) {
135     if (len1 == -1) len1 = this->len;
136     auto g = *this;
137     auto x = g.derivative() * g.inv(len1);
138     x.resize(len1);
139     x = x.integral();
140     x.resize(len1);
141     return x;
142 }
143 Poly exp() {
144     Poly g(1);
145     g.a[0] = 1;
146     for (int l = 1; l < len; l <= 1) {
147         Poly t, g1; t = *this;
148         t.resize(l < 1); t.a[0]++;
149         g1 = (t - g.ln(l < 1)) * g;
150         g1.resize(l < 1);
151         swap(g, g1);
152     }
153     return g;
154 }
155 Poly pow(ll n) {
156     Poly &a = *this;
157     int i = 0;
158     while (i <= a.deg and a.a[i] == 0) i++;
159     if (i and (n > a.deg or n * i > a.deg)) return Poly(a
        .deg + 1);

```

```

160     if (i == a.deg + 1) {
161         Poly res(a.deg + 1);
162         res.a[0] = 1;
163         return res;
164     }
165     Poly b(a.deg - i + 1);
166     int inv1 = ::inv(a.a[i]);
167     FOR (j, 0, b.deg)
168         b.a[j] = a.a[j + i] * inv1 % mod;
169     Poly res1 = (b.ln() * (n % mod)).exp() * (::power(a.a
        [i], n));
170     Poly res2(a.deg + 1);
171     FOR (j, 0, min((ll)(res1.deg), (ll)(a.deg - n * i)))
172         res2.a[j + n * i] = res1.a[j];
173     return res2;
174 }
175 };

```

7.16 Stirling's formula

$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$$

7.17 Theorem

- $1 \sim x$ 質數的數量 $\approx \frac{x}{\ln x}$
- x 的因數的數量 $\approx x^{\frac{1}{3}}$
- x 的質因數的數量 $\approx \log \log x$
- p is a prime number $\Leftrightarrow (p-1)! \equiv -1 \pmod{p}$
- 每個正整數都可以表示成四個整數的平方和
- 任何大於 2 的整數都可以表示成兩個質數的和
- $n^{k-2} \cdot \prod_{i=1}^k s_i$ n 個點、 k 的連通塊、加上 $k-1$ 條邊使得變成一個連通圖的方法數、其中每個連通塊有 s_i 個點

7.18 josephus [0be067]

```

1 // n 個人、每 k 個人就刪除的約瑟夫遊戲
2 int josephus(int n, int k) {
3     if (n == 1)
4         return 0;
5     if (k == 1)
6         return n-1;
7     if (k > n)
8         return (josephus(n-1, k) + k) % n;
9     int cnt = n / k;
10    int res = josephus(n - cnt, k);
11    res -= n % k;
12    if (res < 0)
13        res += n;
14    else
15        res += res / (k - 1);
16    return res;
17 }

```

7.19 二元一次方程式

$$\begin{cases} ax + by = e \\ cx + dy = f \end{cases} = \begin{cases} x = \frac{ed-bf}{ad-bc} \\ y = \frac{af-ec}{ad-bc} \end{cases}$$

若 $x = \frac{0}{0}$ 且 $y = \frac{0}{0}$ ，則代表無限多組解。若 $x = \frac{*}{0}$ 且 $y = \frac{*}{0}$ ，則代表無解。

7.20 數論分塊 [8ccab5]

```

1 // 時間複雜度為  $O(\sqrt{n})$ 
2 // 區間為  $[L, r]$ 
3 for(int i=1 ; i<=n ; i++){
4     int l = i, r = n/(n/i);
5     i = r;
6     ans.push_back(r);
7 }

```

7.21 最大質因數 [ca5e52]

```

1 void max_fac(int n, int &ret){
2     if (n<=ret || n<2) return;
3     if (isprime(n)){
4         ret = max(ret, n);
5         return;
6     }
7
8     int p = Pollard_Rho(n);
9     max_fac(p, ret), max_fac(n/p, ret);
10 }

```

7.22 歐拉公式 [85f3b1]

```

1 // phi(n) = 小於 n 並與 n 互質的正整數數量。
2 //  $O(\sqrt{n})$  · 回傳 phi(n)
3 int phi(int n){
4     int ret = n;
5
6     for (int i=2 ; i*i<=n ; i++){
7         if (n%i==0){
8             while (n%i==0) n /= i;
9             ret = ret*(i-1)/i;
10        }
11    }
12    if (n>1) ret = ret*(n-1)/n;
13
14    return ret;
15 }
16
17 //  $O(n \log n)$  · 回傳 1~n 的 phi 值
18 vector<int> phi_1_to_n(int n){
19     vector<int> phi(n+1);
20     phi[0]=0;
21     phi[1]=1;
22
23     for (int i=2 ; i<=n ; i++){
24         phi[i]=i-1;
25     }
26
27     for (int i=2 ; i<=n ; i++){
28         for (int j=2*i ; j<=n ; j+=i){ // 枚舉所有倍數
29             phi[j]-=phi[i];
30         }
31     }
32
33     return phi;
34 }

```

7.23 歐拉定理

若 a, m 互質，則：

$$a^n \equiv a^{n \bmod \varphi(m)} \pmod{m}$$

若 a, m 不互質，則：

$$a^n \equiv a^{\varphi(m) + [n \bmod \varphi(m)]} \pmod{m}$$

7.24 錯排公式

錯排公式：(n 個人中，每個人皆不再原來位置的組合數)

$$dp_i = \begin{cases} 1 & i = 0 \\ 0 & i = 1 \\ (i-1)(dp_{i-1} + dp_{i-2}) & \text{otherwise} \end{cases}$$

8 String

8.1 AC automation [018290]

```
1 struct ACAutomation{
2     vector<vector<int>> go;
3     vector<int> fail, match, pos;
4     int sz = 0; // 有效節點為 [0, sz] · 開陣列的時候要小心 !!!
5
6     ACAutomation(int n) : go(n, vector<int>(26)), fail(n), match(n) {}
7
8     void add(string s){
9         int now = 0;
10        for (char c : s){
11            if (!go[now][c-'a']) go[now][c-'a'] = ++sz;
12            now = go[now][c-'a'];
13        }
14        pos.push_back(now);
15    }
16
17    void build(){
18        queue<int> que;
19        for (int i=0 ; i<26 ; i++){
20            if (go[0][i]) que.push(go[0][i]);
21        }
22        while (que.size()){
23            int u = que.front();
24            que.pop();
25            for (int i=0 ; i<26 ; i++){
26                if (go[u][i]){
27                    fail[go[u][i]] = go[fail[u]][i];
28                    que.push(go[u][i]);
29                }else go[u][i] = go[fail[u]][i];
30            }
31        }
32    }
33
34    // counting pattern
35    void buildMatch(string &s){
36        int now = 0;
37        for (char c : s){
38            now = go[now][c-'a'];
39            match[now]++;
40        }
41
42        vector<int> in(sz+1), que;
43        for (int i=1 ; i<=sz ; i++) in[fail[i]]++;
44        for (int i=1 ; i<=sz ; i++) if (in[i]==0) que.push_back(i);
```

```
45        for (int i=0 ; i<que.size() ; i++){
46            int now = que[i];
47            match[fail[now]] += match[now];
48            if (--in[fail[now]]==0) que.push_back(fail[now]);
49        }
50    }
51};
```

8.2 Enumerate Runs [94ca46]

```
1 vector<array<int, 3>> enumerate_run(string s){
2
3     int n = s.size();
4     SuffixArray sa(s), saBar(string(s.rbegin(), s.rend()));
5     sa.init_lcp(), saBar.init_lcp();
6
7     set<pair<int, int>> ss;
8     vector<array<int, 3>> runs;
9
10    for (int len=1 ; len<=n ; len++){
11        vector<int> lcp;
12        for (int i=0 ; i+len<n ; i+=len){
13            int pos1 = sa.pos[i];
14            int pos2 = sa.pos[i+len];
15            lcp.push_back(sa.get_lcp(pos1, pos2));
16        }
17
18        for (int ll=0, rr=0 ; ll<lcp.size() ; rr++, ll=rr){
19            while (rr<lcp.size() && lcp[rr]>=len) rr++;
20
21            int preLen = 0;
22            if (ll!=0){
23                int p = n-1;
24                int pos1 = saBar.pos[p-(ll*len-1)];
25                int pos2 = saBar.pos[p-((ll+1)*len-1)];
26                preLen = saBar.get_lcp(pos1, pos2);
27            }
28            int sufLen = rr<lcp.size() ? lcp[rr] : 0;
29
30            int ansL = ll*len-preLen, ansR = (rr+1)*len-1+sufLen;
31            if (ansL!=ansR && ansR-ansL+1>=2*len && ss.find({ansL, ansR+1})==ss.end()){
32                ss.insert({ansL, ansR+1});
33                runs.push_back({len, ansL, ansR+1});
34            }
35        }
36    }
37    return runs;
38}
39
40}
```

8.3 Hash [942f42]

```
1 mt19937 seed(chrono::steady_clock::now().time_since_epoch().count());
2 int rng(int l, int r){
3     return uniform_int_distribution<int>(l, r)(seed);
4 }
5 int A = rng(1e5, 8e8);
6 const int B = 1e9+7;
7
8 // 2f6192
9 struct RollingHash{
```

```
10     vector<int> Pow, Pre;
11     RollingHash(string s = ""){
12         Pow.resize(s.size());
13         Pre.resize(s.size());
14
15         for (int i=0 ; i<s.size() ; i++){
16             if (i==0){
17                 Pow[i] = 1;
18                 Pre[i] = s[i];
19             }else{
20                 Pow[i] = Pow[i-1]*A%B;
21                 Pre[i] = (Pre[i-1]*A+s[i])%B;
22             }
23         }
24
25         return;
26     }
27
28     int get(int l, int r){ // 取得 [l, r] 的數值
29         if (l==0) return Pre[r];
30         int res = (Pre[r]-Pre[l-1]*Pow[r-l+1])%B;
31         if (res<0) res += B;
32         return res;
33     }
34};
```

8.4 KMP [7b95d6]

```
1 // KMP[i] = s[0...i] 的最長共同前後綴長度 · KMP[KMP[i]-1] 可以跳 fail Link
2 vector<int> KMP(string &s){
3     vector<int> ret(n);
4     for (int i=1 ; i<s.size() ; i++){
5         int j = ret[i-1];
6         while (j && s[i]!=s[j]) j = ret[j-1];
7         ret[i] = j + (s[i]==s[j]);
8     }
9     return ret;
10 }
```

8.5 Manacher [9a4b4d]

```
1 string Manacher(string str) {
2     string tmp = "$#";
3     for(char i : str) {
4         tmp += i;
5         tmp += '#';
6     }
7
8     vector<int> p(tmp.size(), 0);
9     int mx = 0, id = 0, len = 0, center = 0;
10    for(int i=1 ; i<(int)tmp.size() ; i++){
11        p[i] = mx > i ? min(p[id*2-i], mx-i) : 1;
12
13        while(tmp[i+p[i]] == tmp[i-p[i]]) p[i]++;
14        if(mx<i+p[i]) mx = i+p[i], id = i;
15        if(len<p[i]) len = p[i], center = i;
16    }
17    return str.substr((center-len)/2, len-1);
18 }
```

8.6 Min Rotation [b24786]

```

1 int minRotation(string s) {
2     int a = 0, n = s.size();
3     s += s;
4
5     for (int b=0 ; b<n ; b++){
6         for (int k=0 ; k<n ; k++){
7             if (a+k==b || s[a+k]<s[b+k]){
8                 b += max(0LL, k-1);
9                 break;
10            }
11            if (s[a+k]>s[b+k]){
12                a = b;
13                break;
14            }
15        }
16    }
17
18    return a;
19 }

```

8.7 Suffix Array [96d6fc]

```

1 // 注意·當 |s|=1 時·lcp 不會有值·務必測試 |s|=1 的 case
2 struct SuffixArray {
3     string s;
4     vector<int> sa, lcp;
5
6     // 0eea96
7     template<typename T>
8     void sais(const T s, int n, int *sa, int *lar, int *p,
9              int *t, int A) {
10         int *rnk = p + n, *q = p + n / 2, *bkt = lar + A;
11         int m = 0, i, j, x = t[n - 1] = 1, y = rnk[0] = -1,
12             cnt = -1;
13
14         for (i=n-2 ; ~i ; i--) t[i] = (s[i] == s[i + 1] ? t[i
15             + 1] : s[i] < s[i + 1]);
16         for (i=1 ; i<n ; i++) rnk[i] = t[i] && !t[i-1] ? (p[m
17             ]=i, m++) : -1;
18         fill_n(lar, A, 0);
19         for (i=0 ; i<n ; i++) ++lar[s[i]];
20         for (i=1 ; i<A ; i++) lar[i] += lar[i-1];
21         auto pushS = [&](int x) { sa[--bkt[s[x]]] = x; };
22         auto pushL = [&](int x) { sa[bkt[s[x]]++] = x; };
23         auto induce_sort = [&](int* v) {
24             fill_n(sa, n, 0);
25             copy_n(lar, A, bkt);
26             for (i=m-1 ; ~i ; i--) pushS(v[i]);
27             copy_n(lar, A - 1, bkt + 1);
28             for (i=0 ; i<n ; i++) if (sa[i] && !t[sa[i] - 1])
29                 pushL(sa[i] - 1);
30             copy_n(lar, A, bkt);
31             for (i=n-1 ; ~i ; i--) if (sa[i] && t[sa[i] - 1])
32                 pushS(sa[i] - 1);
33         };
34         induce_sort(p);
35         for (i=0 ; i<n ; i++){
36             if (~x=rnk[sa[i]]){
37                 j = y < 0 || memcmp(s + p[x], s + p[y], (p[x
38                     + 1] - p[x]) * sizeof(s[0]));
39                 q[y = x] = cnt += j;
40             }
41         }
42     }
43 }

```

```

35     if (cnt+1<m) sais(q, m, sa, bkt, rnk, t + n, cnt + 1)
36     ;
37     else for (i=0 ; i<m ; i++) sa[q[i]] = i;
38     for (i=0 ; i<m ; i++) q[i] = p[sa[i]];
39     induce_sort(q);
40 }
41
42 // 8c731c, lim 要調整成字元集大小·_s 不可以有 0
43 template<typename T>
44 SuffixArray(const T& _s, int lim = 256) : s(_s) {
45     s.push_back(0);
46     int n = s.length();
47     sa.resize(n);
48     lcp.resize(n);
49     vector<int> bkt(n + lim * 2), p(n * 2), t(n * 2),
50         rank(n);
51     sais(&s[0], n, &sa[0], &bkt[0], &p[0], &t[0], lim);
52
53     for (int i=1 ; i<n ; i++) rank[sa[i]] = i;
54     for (int i=0, j, k=0 ; i<n-1 ; lcp[rank[i++]]=k){
55         for (k && k--, j=sa[rank[i]-1] ; i+k<s.size() &&
56             j+k<s.size() && s[i+k]==s[j+k] ; k++);
57     }
58
59     sa.erase(sa.begin());
60     lcp.erase(lcp.begin(), lcp.begin()+2);
61     s.pop_back();
62 }
63
64 // f49583
65 vector<int> pos; // pos[i] = i 這個值在 pos 的哪個地方
66 SparseTable st;
67 void init_lcp(){
68     pos.resize(sa.size());
69     for (int i=0 ; i<sa.size() ; i++){
70         pos[sa[i]] = i;
71     }
72     if (lcp.size()){
73         st.build(lcp);
74     }
75 }
76
77 // 用之前記得 init
78 // 查詢「sa 上的位置」的 x 跟 y 的 lcp
79 int get_lcp(int x, int y){
80     if (x==y) return s.size()-x;
81     if (x>y) swap(x, y);
82     return st.query(x, y);
83 }
84
85 // 回傳 [l1, r1] 跟 [l2, r2] 的 lcp·0-based
86 int get_lcp(int l1, int r1, int l2, int r2){
87     int pos_1 = pos[l1], len_1 = r1-l1+1;
88     int pos_2 = pos[l2], len_2 = r2-l2+1;
89     if (pos_1>pos_2){
90         swap(pos_1, pos_2);
91         swap(len_1, len_2);
92     }
93
94     if (l1==l2){
95         return min(len_1, len_2);
96     }else{
97         return min({st.query(pos_1, pos_2), len_1, len_2
98             });
99     }
100 }

```

```

101 }
102
103 // 檢查 [l1, r1] 跟 [l2, r2] 的大小關係·0-based
104 // 如果前者小於後者·就回傳 <0·相等就回傳 =0·否則回傳
105 // >0
106 // 5b8db0
107 int substring_cmp(int l1, int r1, int l2, int r2){
108     int len_1 = r1-l1+1;
109     int len_2 = r2-l2+1;
110     int res = get_lcp(l1, r1, l2, r2);
111
112     if (res<len_1 && res<len_2){
113         return s[l1+res]-s[l2+res];
114     }else if (len_1==res && len_2==res){
115         return 0;
116     }else{
117         return len_1==res ? -1 : 1;
118     }
119 }
120
121 // 對於位置在 <=p 的後綴·找離他左邊/右邊最接近位置 >p 的
122 // 後綴的 lcp·0-based
123 // pre[i] = s[i] 離他左邊最接近位置 >p 的後綴的 lcp·0-
124 // based
125 // suf[i] = s[i] 離他右邊最接近位置 >p 的後綴的 lcp·0-
126 // based
127 // da12fa
128 pair<vector<int>, vector<int>> get_left_and_right_lcp(int
129     p){
130     vector<int> pre(p+1);
131     vector<int> suf(p+1);
132
133     { // build pre
134         int now = 0;
135         for (int i=0 ; i<s.size() ; i++){
136             if (sa[i]<=p){
137                 pre[sa[i]] = now;
138                 if (i<lcp.size()) now = min(now, lcp[i]);
139             }else{
140                 if (i<lcp.size()) now = lcp[i];
141             }
142         }
143     }
144     { // build suf
145         int now = 0;
146         for (int i=s.size()-1 ; i>=0 ; i--){
147             if (sa[i]<=p){
148                 suf[sa[i]] = now;
149                 if (i-1>=0) now = min(now, lcp[i-1]);
150             }else{
151                 if (i-1>=0) now = lcp[i-1];
152             }
153         }
154     }
155     return {pre, suf};
156 }
157
158 };

```

8.8 Wildcard Matching [e93074]

```

1 const int B = 27;
2 string p;
3

```

```

4 | for (int i=0 ; i<B ; i++) p += ('a'+i);
5 | p += '*';
6 | vector<vector<double>> res(B);
7 | for (int i=0 ; i<B ; i++){
8 |     vector<double> ss, tt;
9 |     for (auto x : s) ss.push_back(x==p[i] || x=='*');
10 |    for (auto x : t) tt.push_back(x==p[i] || x=='*');
11 |    reverse(tt.begin(), tt.end());
12 |    res[i] = PolyMul(ss, tt);
13 | }
14 | for (int i=t.size()-1 ; i+t.size()-1<res[0].size() ; i++){
15 |     int total = 0;
16 |     for (int j=0 ; j<B-1 ; j++) total += (int)abs(round(res[j
17 |         ][i]));
18 |     total -= 25*(int)abs(round(res[26][i]));
19 |     cout << (total==t.size());
20 | }

```

8.9 Z Algorithm [9d559a]

```

1 | // z[i] 回傳 s[0...] 跟 s[i...] 的 lcp, z[0] = 0
2 | vector<int> z_function(string s){
3 |     vector<int> z(s.size());
4 |     int l = -1, r = -1;
5 |     for (int i=1 ; i<s.size() ; i++){
6 |         z[i] = i>=r ? 0 : min(r-i, z[i-l]);
7 |         while (i+z[i]<s.size() && s[i+z[i]]==s[z[i]]) z[i]++;
8 |         if (i+z[i]>r) l=i, r=i+z[i];
9 |     }
10 |     return z;
11 | }

```